

Non-Degenerate Construction and the Kernel of Admissibility

An Exhaustion Theorem for Standing, Reference, and Irreversibility

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Abstract. Actual non-degenerate determinate objecthood necessarily instantiates Admissibility, Standing, Reference, and Irreversibility. This paper defends that claim in two stages: a constitutive argument identifies the commitments incurred by an actual identity-bearing object or incidence, and a reconstruction establishes the four functional roles required to satisfy them. The necessity argument has an explicit mathematical development: original incidence supplies canonical qualification, and exact realization is equivalent to preservation of the original observation profile on every encoding fibre. Omission, surplus, and erasure give witnessed failure criteria. Physical qualification belongs to the incidence and its original conditions, including when nobody observes it; descriptions are accountable to those conditions. The kernel therefore governs determinate physical and mathematical realization by necessity. A representation-conservation theorem then establishes agreement of every independently warranted answer about the same bearer and complete query. Its proof uses determinacy of the relevant result incidence and the separate warrant of each answer. Agreement between representations is the conclusion. The theorem applies to arbitrary families of governed queries and yields an exclusion principle: a proposed additional status either contributes no governance difference, changes the bearer or constructional question, or lacks standing as an answer to the unchanged assessment. A changed construction remains kernel-governed; an inaccurate representation fails its answer-role without thereby changing its subject. Binary admission, exact contextual quotients, and preservation of partial-operation domains are subsequently distinguished and connected. Functional minimality and finite authorization-boundary analysis establish the core exhaustion claim. Technical appendices develop exact representation criteria, structural consequences, and selected mathematical results checked in Lean, including the constructed semantic necessity argument, with its original-domain interpretation and proof premises explicitly stated.

Keywords: admissibility; standing; reference; irreversibility; determinate objecthood; functional necessity; representation conservation; contextual

equivalence; partial operations; bivalence; AMetric boundary; faithful representation

Scope. Kernel governance is necessary for every instance of non-degenerate determinate objecthood, including unobserved physical incidences. The necessity concerns actual realization and holds independently of its description. Representation conservation fixes the bearer, complete question, and original conditions independently, and requires determinacy of the particular result relation. Admission for that specified use is binary; the complete object retains its substantive properties and their original domains of use.

1. The foundational claim and its proof order

An actual physical incidence has its relata and conditions whether or not anyone observes it. Whether a particular arrangement carries a load is settled by that arrangement's physical relations under those conditions. A record has the relation to an occurrence that its actual formation supplies. Identifying, describing, or using either is answerable to the incidence concerned. The foundational question is what such actual determination necessarily involves.

“Determinate” here concerns an object or incidence that can remain the same identified subject across a faithful change of presentation. “Non-degenerate” requires an actual identity-bearing relation or use, so that the determination is realized rather than merely attributed by a label. In the physical case, the relation can obtain without an act of assessment. The argument begins with this actual subject matter and examines what would be lost if its required distinctions were removed.

The kernel consists of four functional roles:

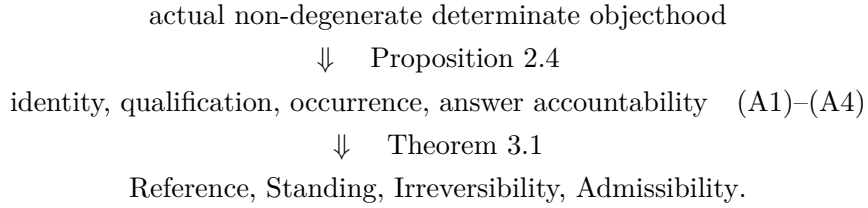
$$K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}.$$

Reference fixes the actual bearer, relata, and relevant conditions of sameness. Standing is the construction's actual qualification for the specified relation or use. Irreversibility retains which occurrence instantiated which conditions; a later occurrence cannot substitute for it. Admissibility governs the entry, transformations, and combinations under which those conditions remain jointly satisfied. An inspector may discover or misreport a qualification; endorsement does not confer it. The roles can

be implemented implicitly or together. Their necessity concerns the work of determinate realization itself.

Any physical domain containing non-degenerate determinate objects, states, carriers, records, or endpoints therefore instantiates these roles by necessity. This includes incidences no observer assesses. The kernel is not an optional interpretive overlay on independently complete physics: determinate physical realization already incurs the work, and its descriptions inherit the corresponding obligations. The domain supplies its substantive relations and laws. The kernel identifies what their determinate instantiation must accomplish; its reconstruction uses no metric quantity or particular dynamical law.

The two necessity steps. The proof separates the claim about actual objecthood from the reconstruction of its functions:



Proposition 2.4 defends why actual identity-bearing determination incurs each commitment. Theorem 3.1 then reconstructs the functions that satisfy them. Defining a realization to be adequate would not establish the first step. The later conservation theorem has its own explicit grounds: independently identified bearer and complete query, determinacy of the particular result relation, and each answer’s original warrant. Thus role necessity and warranted representational agreement have distinct, inspectable justifications.

Between these steps, Proposition 2.6, Lemma 2.7, Theorem 2.8, and Corollary 2.9 construct the semantic assessment, characterize retention of original identity and observations, and give witnesses against omission, surplus, and loss of recoverable work.

“Minimal” means that genuinely removing a function loses an indispensable condition. A replacement mechanism doing the same work instantiates that function. The argument does not require four separate components or derive a domain’s substantive law from the four role names.

1.1. One familiar construction: a bridge and a toy car

Consider a small bridge built across a gap. The task is fixed before testing: this bridge must carry this toy car across this gap without anyone holding it up. In the first trial, a loose joint gives way and the bridge collapses. After the joint is repaired, the car crosses successfully in a second trial. These are actual physical occurrences; their relations obtain or fail independently of an inspector. The collapse remains a determinate event, while the original arrangement fails to qualify for the specified successful crossing. Both trials necessarily instantiate the kernel. The four roles identify the work needed to retain those occurrences and qualifications.

Reference: what is the claim about? The bridge, car, gap, configuration, and occurrence of testing must remain identifiable. Demonstrating a different bridge, using a lighter car, or shortening the gap can establish a different result. Counting it as the original demonstration loses the fixed subject of assessment. The failure is substitution: an answer about one construction is made to stand for an answer about another.

Standing: does the construction qualify for this use? For the task as stated, an assembly's bridge-like appearance and a label saying "passed" cannot confer the qualification supplied by actually carrying the car under the required conditions. Removing this distinction loses the connection between the claimed achievement and the assembly's qualification for that use. Standing concerns that actual qualification. Discovering it and reporting it are further acts, whose correctness must answer to the construction.

Admissibility: are the conditions of the construction respected? Suppose someone supports the bridge with a finger while the car crosses. The crossing occurs, but the finger changes what bears the load. The assisted arrangement can succeed as that different support system. Counting its success as the bridge carrying the car without outside support discards a condition of the original task. Admissibility governs the conditions on construction, combination, and use; standing concerns the qualification of the particular construction at the specified use. A mechanism may perform both functions. Renaming the extra support does not remove its physical contribution.

Irreversibility: which occurrence succeeded? The repaired arrangement can succeed in the second trial. That success does not make the first trial successful under its original conditions. Removing the distinction substitutes a later occurrence for the original one. The required fixation concerns which incidence occurred, independently of whether evidence of it survives. Repair and learning remain possible, with each result attributable to its own occurrence and conditions.

The task selects the physical relation being assessed; it does not make that relation depend on an inspector's permission. Once the bridge, car, gap, support conditions, and trial are fixed, a label cannot alter whether the car was carried by that arrangement. These examples expose different losses of constructional work, and several can occur together. Another mechanism preserving the missing work would continue to instantiate the corresponding role.

The complete construction is fixed before comparison. For the original performance, the identified subject includes the arrangement, occurrence, load, and support conditions. A repair changing those data presents a different constructional performance. A supported crossing answers a different question from an unsupported one. Neither supplies a contrary answer about the unchanged original trial. A false report that continues to claim that original trial instead lacks standing as an answer there. These cases introduce no additional standing state of the unchanged complete construction. Every new determinate arrangement, occurrence, or inscription remains necessarily kernel-governed. Sections 2 and 3 give the general argument.

1.2. The central conservation argument

The representation problem is often obscured by treating a proposed encoding as though it already had standing as a presentation of its target. A data structure can store a label without establishing that the label correctly describes anything. An arbitrary rule operating on that structure can distinguish spellings without distinguishing the standing of the represented object. The appropriate universal statement concerns every admissible presentation of the same complete object and every warranted answer to the same complete question.

The argument is direct. Fix a bearer and a question before comparing answers. Specify, independently of the encodings, the relation under which

an answer is warranted for that bearer and question. If the assessment admits at most one warranted result, any two answers warranted by that relation agree. Therefore two representations that retain the same bearer and question and independently warrant their outputs agree. Conversely, discordant answers cannot both retain that standing at the unchanged assessment. Agreement is the conclusion; the premises identify each answer's own warrant.

This theorem is not restricted to a finite syntax of observations. It applies to any family of determinate governed queries. Computable interfaces admit an additional compositional test: primitive preservation of domains and outputs entails preservation of all generated finite contexts. The semantic theorem establishes conservation for arbitrary admissible presentations; the primitive theorem supplies a method for verifying an implementation.

1.3. Three distinctions required by the theorem

The object and its answer. The bearer is identified through its original occurrence, material, target, and conditions. Agreement of answers cannot be used retrospectively to define numerical identity. A later discovery may correct a mistaken assessment of unchanged data. A later construction may supply different data. The distinction between these cases is part of historical accountability, not a prohibition on correction or learning.

Standing and a complete profile. For one specified constructional use, admission is binary. The actual status at one complete determinate assessment is unique. Across different assessments, both polarities may be realized. A target, proof term, diagnostic record, or compatibility condition remains substantive data wherever an original observation reads it. Two constructions need not be the same merely because both are admitted. Accordingly, representation conservation and binary compression of an entire interface are connected by an explicit semantic realization theorem, rather than identified by terminology.

A representation and a claim to represent. A sound representation supplies independently warranted answers about its identified bearer. A purported redescription that changes a standing-bearing answer while

claiming the same bearer and question fails that claim. The new inscription may itself be a determinate object; that does not confer standing on its use as a representation of the original one. This distinction prevents an inadmissible substitution from being counted as a third state of the unchanged object.

1.4. Contribution and relation to existing work

Individuation and persistence. Wiggins’s account of individuation connects identity questions with the kind of thing concerned while defending the absoluteness of identity [1, chs. 1–3]. The connection matters here because identifying the same bridge, physical occurrence, or derivation requires more than retaining a name. The present argument leaves substantive criteria of individuation and persistence to the domain. Its further claim concerns the work necessarily incurred when those criteria determine an object or incidence. A domain can identify a continuing physical artifact through a sequence of changes. That persistence question differs from the identity of the complete arrangement and trial used in the original assessment. Persistence of the artifact does not establish that changed material, conditions, or occurrences are admissible redescriptions of the original complete performance. The relevant identity criteria are fixed before its results are compared.

The challenge from transcendental arguments. Stroud’s criticism makes a necessary methodological demand: conditions under which we must think or speak of a world do not, without further argument, establish how that world must be [2, pp. 255–256]. Proposition 2.4 therefore begins with actual non-degenerate determinate incidence, not merely a subject’s belief that an incidence exists. Its question is what the occurrence and its relations must accomplish to be the determinate occurrence concerned. Whether a particular arrangement bears a load, or a record stands in the claimed relation to an event, is fixed by that arrangement or relation even if nobody assesses it. The foundation must defend the passage from such identity-bearing incidence to qualified use; calling the argument “constitutive” would not supply that defense. Section 2 gives it and identifies the domain-specific conditions it leaves to physics or mathematics.

Representation and uniqueness. Luce and Suppes distinguish the existence of a measurement representation from the uniqueness of representations under transformations of scale [3, pp. 2–5]. Their distinction between equal measured weight and identity of the measured objects is equally relevant. Agreement of answers to one question does not determine numerical identity, and an encoding does not become faithful merely because it assigns values. The present fixed-contract analysis follows the same discipline of specifying which relations and answers a representation preserves. Its query families also include qualification for use, original occurrence, contextual behavior, and partial definedness. The resulting uniqueness is over the same complete contract; it neither identifies objects with equal measurements nor derives a unique substantive policy from the four functional roles. Measurement representation supplies a close mathematical comparison, while the necessity of those roles requires the separate foundational argument.

Proof identity. Došen distinguishes normalization-based and generality-based criteria for identifying proofs [4, secs. 2–3]. His discussion of the two projections $p \wedge p \rightarrow p$ makes the issue concrete: the derivations have the same premise and conclusion, but generalizing their formula occurrences can distinguish the component selected. Thus equal conclusions do not establish equal proofs. Here the full-profile quotient identifies presentations only when every observation in the justified complete contract agrees. A contract recording only conclusions can be coarser than one that observes the proof’s structure. The conservation theorem begins with an independently established common bearer and independently warranted answers; it does not infer the sameness of proof objects from their common conclusion. Its observational equivalence therefore answers a specified representation problem without purporting to settle every philosophical criterion of proof identity.

Justification and checkable derivation. Martin-Löf explains implication elimination through the verification supplied by its premises, connecting its justification with proof reduction [5, pp. 45–48]. This provides a direct comparison for the distinction between producing a formal result and warranting its use. Rewriting theory and proof assistants make precise the corresponding questions about reduction and checkable judgments [6, 7]. In the worked checker example of Section 6, soundness is derived from separately stated inference rules, rather than assumed as a certificate field.

That proof settles the checker’s relation to those rules. The justification of the rules for an intended use, and the foundational necessity of the kernel, remain distinct arguments. None of the cited accounts is invoked as a proof of the four-role thesis.

The contribution is the resulting proof architecture: a necessity argument for the governance of determinate objecthood, a conservation theorem grounded in individual warrants, and exact representation consequences with their premises exposed. The mathematics—functionality, quotient factorization, partial-operation congruence, and finite dependency arguments—is used to separate those burdens.¹

1.5. Proof order and grounds for objection

Section 2 states and defends the objecthood commitments; Section 3 recovers the four roles and their functional minimality. Sections 4 and 5 establish the authorization boundary and distinguish role necessity from exact agreement under a complete contract. Section 6 derives representation conservation from independent incidence warrants. The principal result and its dependency structure are then assembled in Section 7. Section 8 gives the compact account of binary admission and the conditions for a binary effective quotient, followed by the interpretations in Section 9. Appendices A, B, and C supply the detailed descent proofs, quotient structure, and mechanization scope.

Replacing the names of the roles while preserving their work is not an omission. This observation does not settle every objection: the constitutive analysis of objecthood can be challenged, an inference can fail, or a proposed application can misidentify its bearer, omit a relevant question, or lack determinate and independently warranted answers. Such objections address the actual foundations, deductions, or application conditions. The paper’s organization makes those burdens separately assessable.

¹ Companion manuscripts in AASC (Admissibility And Standing Constraint) develop downstream applications to bivalence, coherent trajectories, role occupancy, and anchor–tensor–skin analysis [10, 11, 12, 13]. They are not premises of the foundational or conservation arguments established here.

2. Determinate Objects and the Conditions of Construction

The foundational question is what must already be true of an actual construction or incidence for its determination to be realized. “Determinate” concerns the same identity-bearing subject across a faithful change of presentation. “Non-degenerate” requires at least one actual identity-bearing relation or successful use. A physical incidence can supply such a relation without a description, observer, or certification event. Marks merely purporting to describe an object do not by themselves supply the relation to that object.

In what follows, *qualification* concerns whether the actual bearer satisfies the original conditions of a specified relation or use. *Assessment* names that question and its settled answer on the evaluated domain; discovering the answer is a further epistemic act. *Warrant* concerns the original grounds for relying on that incidence in the claimed role. For physical construction, these grounds are the actual relations and conditions; for inference, they are the premises and rules that license the step. These uses of the terms require neither a human authorizer nor a computable decision procedure.

The observation frame below states the questions before their verdicts. The adequacy clauses then make the proposed requirements explicit. Proposition 2.4 supplies the substantive argument that actual non-degenerate determination incurs them; the definition of adequacy does not substitute for that argument.

2.1. Questions and occurrence identity before verdicts

DEFINITION 2.1 (Base observation frame). A base observation frame is

$$\mathcal{B}_D = (E_D, T_D, U_D, H_D, \mathcal{F}_D, \mathcal{R}_D, \mathcal{Q}_D).$$

Here E_D consists of candidate performance records, T_D of proposed targets, U_D of specified uses, and H_D of finite retained histories. The families $\mathcal{F}_D$ and $\mathcal{R}_D$ name candidate construction contexts and candidate changes of presentation. Their declaration does not certify their definedness or fidelity. The observation questions $\mathcal{Q}_D$ ask which target is concerned, whether a specified use is warranted, which occurrence and original conditions a historical claim concerns, and whether these answers survive the declared contexts. Definedness questions and the contextual

consequences relevant to the task are included. The family of questions need not be finite or generated by a finite observation language.

A question can be declared before its answer is supplied. A proof-checking problem can identify a proposed conclusion and the question whether a displayed transition licenses it without deciding that question. Likewise, comparing two encodings does not first require declaring them equivalent. This order matters: comparison positions must not be selected retrospectively to erase a difference that an alleged equivalence cannot preserve. The frame fixes what is being asked, with sufficient contextual detail to distinguish different uses; it does not prescribe a validator, a complete profile of answers, or the cardinality of a later quotient.

DEFINITION 2.2 (Occurrence and act identity). A performance record specifies an occurrence, the material performed, its original context, and the target claim under assessment. Two records represent the same act only under an identification preserving that occurrence and those original data. The verdict about the act is not a coordinate defining its numerical identity. Changing the performed material or original context supplies a different performance or sequence; learning something further about unchanged data supplies a new assessment of the original performance.

Occurrence identity is therefore fixed before the success question is answered. Retaining an occurrence in a faithful account means retaining which occurrence is concerned; the occurrence need not leave an accessible record, and its original facts need not remain recoverable. Records may be incomplete, and the evaluated fragment may consequently be partial. A notation for a record also need not be unique: recoding an occurrence identifier can preserve its bearer. What is excluded is using equality of verdicts to decide retrospectively which occurrence a verdict concerned. In an atemporal construction, the relevant occurrence is a position or instantiation in the construction; no independent physical arrow of time is assumed.

2.2. Target adequacy

DEFINITION 2.3 (Target-adequate realization). A realization of $\mathcal{B}_D$ is target-adequate when its evaluated fragment is nonempty, at least one specified constructional use is warranted, and it satisfies the following requirements.

- (A1) *Target determinacy.* Every evaluated performance concerns a repeatable target. A change of presentation claimed to preserve that performance preserves the target under the declared identification.
- (A2) *Assessment and use.* For every evaluated performance/use query under complete stated conditions, the realization settles whether that use is warranted by the performance under those conditions. A favorable assessment supports that use under those conditions; an unfavorable assessment cannot support it as though favorable. A different use or different conditions present a different assessment question.
- (A3) *Historical fixation.* Within a retained history, the original occurrence, original conditions, and target of its assessment remain identifiable. A correction can produce a further performance or alter the conditions of a new assessment. It cannot substitute different original data while reporting that the original historical performance succeeded on its original data.
- (A4) *Answer accountability.* Every answer offered with standing by a purported same-object redescription or encoding must be warranted for the originally identified bearer and complete question. Warrant concerning a different bearer, use, or set of conditions cannot be transferred merely by retaining a label. This is a requirement on each answer separately; agreement between representations will be derived from their common assessment semantics.

These clauses specify a task rather than a four-component implementation. In particular, (A2) supplies the assessment-to-use connection: producing a score or predicting another system’s verdict does not, merely as such, establish what may be done with the assessed performance. “Warranted” is used structurally: the use succeeds under the conditions that constitute the target, rather than by replacing the target or its conditions. For a derivation, this concerns licensed inference; for an identity-bearing physical incidence, it concerns which identifications and continuations are true of that incidence. A realization need not be a human decision procedure or a computable test. Its answers may be settled semantically even when no observer can discover them.

Two nonvacuity claims must be distinguished. An actual standing construction supplies a positive instance. A proper rejection boundary additionally requires an actual failed locus. Neither an empty presentation space nor a regime that refuses every constructional use supplies the

positive target of this paper. The existence of a failed locus is stated separately whenever a result needs both polarity values. This does not assert that every fixed assessment question has both positive and negative examples. Nor does it make all observations Boolean: target identities, provenance, and continuation behavior can have many values. Assessment of one fully specified use and the total profile of a performance remain different objects.

PROPOSITION 2.4 (Objecthood incurs the target commitments). *Non-degenerate determinate objecthood, as the identity-bearing subject of construction or incidence, incurs (A1)–(A4). In particular, once a physical object, event, or incidence is established as non-degenerate and determinate, application of these commitments requires no further governance permission.*

PROOF. Begin with an actual non-degenerate determinate object or incidence. Its relata and original conditions are supplied by the domain before the proposed answer or representation. The argument asks what is lost if each required distinction is genuinely removed while this same subject matter is retained.

Identity and redescription (A1). An incidence concerns its actual bearer and relata. A purported preservation that substitutes different relata does not preserve that incidence. If there is no distinction between retaining and substituting its subject, the incidence lacks the claimed determinate identity. Faithful changes of presentation must therefore retain the target under its independently fixed identification. This gives (A1).

Qualification and use (A2). Under its original conditions, an actual incidence determines which specified relations hold of that bearer. Qualification for a use of it depends on those conditions being satisfied for that role. A bridge’s relation to a load and a record’s relation to an occurrence are fixed by their actual realization, irrespective of endorsement. Removing the distinction between satisfying and failing those conditions makes the proposed use independent of the incidence it purports to use. Retaining the incidence as its ground therefore incurs (A2). The assertion concerns evaluated uses under complete conditions; neither unique answers to arbitrary questions nor an observer’s ability to decide them is inferred from object identity.

Original occurrence (A3). An incidence as originally instantiated has those relata under those conditions. A later replacement instan-

tiates different original data or a different occurrence. Removing this distinction identifies the original incidence with its substitution and loses which incidence the historical claim concerns. Any retained history that continues to concern the original must therefore preserve the distinction in (A3). Loss of evidence can limit what is known; it does not make a later occurrence the original one. The same reasoning applies to an instantiation or position in an atemporal mathematical construction.

Answer accountability (A4). A use offered as concerning this bearer under this question must have its grounds there. Grounds concerning a substituted bearer, role, or set of conditions concern a different incidence. Transferring them by retaining a label would remove the identity and qualification distinctions just established. Each answer’s warrant must therefore remain attributable to its original bearer and complete question, as required by (A4). This is a pointwise requirement; pairwise conservation follows later from independently established incidence and result determinacy.

Non-degeneracy supplies an actual identity-bearing instance. In particular, the bearer qualifies for the identification supplied by its actual relation under the original conditions; carrying that identification forward faithfully is a warranted constructional use. This supplies the positive-use requirement of target adequacy without requiring an observer to exercise it. The four clauses express requirements of retaining that determination in its original role; each asserted omission above loses identified work rather than merely a preferred name for it. $\square$

The proposition is a necessity argument about the constitutive commitments of determinate objecthood. Each commitment is traced to the original identity-bearing incidence, and its removal is assessed by what would be lost. The mathematical development below constructs its semantic realization and proves exact criteria for preservation, omission, surplus, and erasure. It is not a set-theoretic inference from the existence of an uninterpreted element $x \in X$. Such an element acquires the relevant significance when it is offered as a determinate object or incidence; even its repeated identification then answers to conditions of correct and incorrect use. The argument is therefore not restricted to an optionally selected class of “kernel-friendly” objects. The qualification is the claim to non-degenerate determination itself. Equally, the proposition does not determine a physical theory, decide empirical identifications, or replace

a domain's substantive conditions of success. It identifies what those conditions must accomplish wherever there is determinate objecthood.

2.3. The mathematical content of the necessity argument

The preceding argument admits a mathematical development without introducing a structure whose fields already assert (A1)–(A4). The primitive data below specify the original subject and its substantive relations. Qualification, its semantic assessment, the limits on transferring a warrant, and the criterion for a successful presentation are derived from those data. No proposed encoding is certified merely by being attached to an actual object.

DEFINITION 2.5 (Original incidence semantics). Let X be the domain of independently identified original objects or occurrences, U the complete specified uses or questions, and V_u their possible result types. The original domain supplies

$$J(x, u, a), \quad x \in X, \quad u \in U, \quad a \in V_u.$$

This means that a is actually incident at x under the original conditions of u . It does not mean that a reader has printed a . At least one actual triple is witnessed for the nonempty target of the proposition. Define qualification for realizing an answer to u by

$$Q_J(x, u) \iff \exists a \in V_u J(x, u, a).$$

If J lists all outcomes of an experiment, Q_J says that an outcome occurs; it does not say that a chosen outcome or practical success condition is realized. For a specified successful use, J must be that use's original witness relation. A substantive use can equivalently be supplied directly by its independently interpreted satisfaction condition $Q(x, u)$. It is represented in this form by taking a one-point result type and setting $J(x, u, *) \leftrightarrow Q(x, u)$.

The carrier X consists of semantic identities at the level of the original claim. Coordinate strings, gauge descriptions, and redundant records belong to a presentation space P , with an independently justified interpretation $\rho : P \rightarrow X$. They need not be different elements of X . Where the original claim concerns a complete performance, its occurrence, arrangement, conditions, and target are identified before its verdict, as

in Definition 2.2. A domain may require a quotient to express that identification. Neither the present construction nor the later quotient theorem licenses choosing that quotient after conflicting answers are seen.

PROPOSITION 2.6 (Intrinsic qualification and semantic assessment). *For every original incidence relation J , an actual instance $J(x, u, a)$ qualifies its bearer for that incidence. A successful positive qualification report at (x, u) requires $Q_J(x, u)$; a settled negative report requires $\forall a \neg J(x, u, a)$. These qualifications are incompatible. In classical logic there is exactly one such Boolean qualification answer at each complete (x, u) , and its value is true exactly when $Q_J(x, u)$ holds. No functionality of J is needed for this qualification result.*

PROOF. The actual result a witnesses the existential defining $Q_J(x, u)$. A positive witness and absence of every such witness cannot coexist. By excluded middle, either a witness exists or none does. Assign true in the first case and false in the second. This gives existence of the canonical qualification answer. Any other answer meeting the same original satisfaction condition has the same polarity, proving uniqueness. An independently specified use condition is covered by the one-point construction in Definition 2.5. $\square$

Qualification is fixed by J before this characteristic function is written. The function records its truth; it does not confer it. The actual-instance and incompatibility arguments are constructive. Classical logic supplies the total Boolean presentation of qualification; this is not a derivation of excluded middle from the kernel, an executable decision procedure, or a claim that every experiment has one physical outcome. A relation may have several incident values while existence of some incident value remains one qualification question. Uniqueness of a particular result is the separate premise of Section 6. An unanswered query is likewise not a proved negative qualification.

LEMMA 2.7 (Identity separation and unconditional warrant transfer). *For original subjects $x, y \in X$, the following are equivalent:*

- (i) $x = y$;
- (ii) for every $z \in X$, $x = z$ if and only if $y = z$;
- (iii) $\forall F : X \rightarrow \text{Prop}, F(x) \Rightarrow F(y)$: every original predicate true of x is true of y , without a further domain premise.

The same assertion applies to complete original claim indices, including their bearer, use, conditions, and proposed result.

PROOF. Equality permits substitution in every predicate, giving (ii) and (iii). In (ii), take $z = x$. For (iii), choose the identity predicate $F(t) \equiv (t = x)$. Its truth at x forces $y = x$. Equivalently, for $x \neq y$ this predicate is a concrete true-at-source, false-at-destination witness against an unconditional transfer rule. $\square$

The quantifier over predicates tests a transfer claimed without a substantive domain justification. It does not prohibit a particular theorem proving that a particular property passes between distinct objects. Such a theorem supplies the further premise. Nor are identity predicates proposed measurements or pre-fixational selectors. They express what a claim to retain this already identified subject means. They are equivariant: under an identity-preserving relabeling, both subject and comparison parameter are relabeled. No preferred occupant is selected.

This gives a precise defense of answer accountability. An incidence at a different ground is not, solely by being an incidence, evidence for the original claim. A successful use must establish the original satisfaction condition or supply a proved relation that entails it. A false report can remain a determinate physical inscription while failing that report-role. Its own occurrence and its incorrect claim are different original questions.

THEOREM 2.8 (Existence and necessity of an exact original realization). *Let $F : X \times A \rightarrow \text{Prop}$ be any independently fixed family of original observations, $\rho : P \rightarrow X$ the original interpretation of presentations, and $e : P \rightarrow C$ a proposed encoding. There exists a reader $R : C \times A \rightarrow \text{Prop}$ satisfying*

$$R(e(p), a) \iff F(\rho(p), a) \quad \text{for all } p, a \quad (\text{N1})$$

if and only if every encoding fibre preserves the original observation profile:

$$e(p) = e(p') \implies (F(\rho(p), a) \leftrightarrow F(\rho(p'), a)) \quad \text{for all } a. \quad (\text{N2})$$

On the realized code image, the exact reader is unique up to agreement on every observation. For the identity family $F(x, z) \equiv x = z$, the criterion is exactly

$$e(p) = e(p') \implies \rho(p) = \rho(p').$$

It requires no injectivity on redundant presentations of the same original subject.

PROOF. If an exact reader exists, equal codes yield the same proposition $R(c, a)$. Applying (N1) twice gives (N2). Conversely, define

$$R(c, a) \iff \exists p (e(p) = c \wedge F(\rho(p), a)).$$

An original observation at p supplies the required witness for $R(e(p), a)$. In the converse direction, any witnessing p' has the same code as p , so the fibre condition transfers its original observation to p . This proves exactness without a choice of representative. Two exact readers agree at any realized code by taking a presentation of that code and applying their equations. The identity specialization follows from Lemma 2.7. $\square$

Existence of one canonical realization alone would not establish necessity. The converse criterion quantifies over every proposed encoding and every possible reader, and supplies an obstruction when original work is genuinely lost. The original domain itself always supplies an exact realization: take $P = C = X$, with $\rho = e$ the identity and $R = F$. This existence statement does not assume an adequacy certificate. The theorem then determines exactly which other encodings can perform the same work. Exact retention of original identity also determines a unique reconstruction of the original subject on the realized code image: assign to a code the original subject of any presentation producing it; the fibre criterion makes this independent of the presentation. Consequently every already specified original predicate is recoverable semantically. This supplies no algorithm for discovering its truth and adds no new domain fact. General observation families preserve their original contextual distinctions; the identity specialization applies when full same-object retention is claimed. Adding identity questions to a task that claims only a coarser observation would change that task and is not licensed by the theorem.

COROLLARY 2.9 (Witnesses against a proposed realization). *A fixed reader fails (N1) exactly when it omits an original true observation or asserts an observation that is false at the original ground. A proposed encoding admits no exact reader exactly when some code fibre merges different original observation profiles. The latter alternative concerns actual distinctions in the original family, not an imposed number of role names or storage fields.*

PROOF. Negating the exact-reader biconditional yields a presentation and observation at which one direction fails: either the original observation

holds and the reader does not, or the reader holds and the original observation does not. Negating the equivalent fibre criterion yields presentations with the same code and a differing original observation. Conversely, either witness contradicts the relevant exactness equation. Classical logic is used to extract witnesses from the negated universal claims; each displayed witness refutes exactness directly. $\square$

A singleton domain, a constant coordinate, or a derived implementation can perform necessary work without storing four independent fields. The criterion therefore tests loss of the function, including its implicit implementation. If deleting a component leaves the original profile recoverable, it has not deleted the relevant work. If it destroys recoverability, the theorem supplies the precise obstruction. In particular, preservation of every required answer alone is insufficient: an additional false answer is a witnessed defect even when none of the true answers is omitted.

How the construction establishes the four commitments. Original subject identity gives the separating tests of Lemma 2.7; a successful full same-object realization must retain them by Theorem 2.8. Applying any independently fixed target map to that identity yields (A1). Original qualification is supplied by the domain relation and has the exact assessment in Proposition 2.6, giving (A2) without observer endorsement. The original occurrence and its conditions are fixed arguments of that relation. A later success at different arguments cannot change a false original qualification; faithful retention of the original argument preserves every predicate on it. This yields (A3) independently of whether a record survives. Finally, Lemma 2.7 excludes warrant transfer justified by subject substitution alone, and Corollary 2.9 identifies any unsupported assertion at the original ground. These are the mathematical forms of (A4).

The premises are the actual original relations and their independently identified subjects, not a field asserting target adequacy, agreement, or kernel governance. Exactness is the claim made by a proposed realization and is characterized and witnessed by the theorem; its soundness direction is not presented as an independently discovered fact about every proposed reader. The ontic direction is maintained throughout: an incidence has its actual qualification independently of whether it is represented, and representation succeeds only by answering to that qualification.

A representation that fails this use remains an object with its own determinate incidence. The domain interpretation remains accountable to what is actually being modeled, as it must for any mathematical account of physical subject matter.

2.4. Does objecthood already require qualified use?

The key passage in Proposition 2.4 is from actual identity-bearing incidence to qualified use in (A2). An unobserved rock or physical event can be determinate without anyone assessing it, using it as a premise, or authorizing a description. An account of correct descriptions alone would therefore leave the physical necessity claim unestablished. A necessity of thought or description does not, merely as such, establish a necessity in its objects; that is a familiar pressure on transcendental arguments [2]. If the kernel only prescribed how an investigator should speak, it would not establish the claimed necessity for physical objecthood.

The reply starts from an actual determinate incidence, rather than inferring its existence from the requirements of a description. Consider the bridge, car, support arrangement, and trial in Section 1.1. Whether that arrangement carries the car depends on the actual physical relation under those conditions. The relation has those relata, rather than a substituted bridge or load; it obtains or fails for that arrangement and occurrence, rather than by an observer's endorsement. A task can select this relation for assessment, but does not create its physical satisfaction conditions. The distinction between the incidence and a substitution remains whether anyone records it or not. Conditions of individuation and reidentification, rather than the mere persistence of a name, are at issue here; compare the discussion of identity and sortal concepts in [1].

The physical example makes the general deletion argument inspectable. Its relata fix what actually occurs; qualification distinguishes the relation that obtains from one that would require different conditions; admissible combination retains those qualifications together; original occurrence distinguishes this incidence from a replacement. The constructional work belongs to the incidence independently of its description. Theorem 3.1 identifies these functions as Reference, Standing, Admissibility, and Irreversibility. Their implementation can coincide. A later repair changes the arrangement or occurrence assessed, whereas later evidence can improve knowledge of the original one.

The failed bridge and its collapse remain determinate physical construction. The collapse instantiates its own relations even though the attempted crossing does not qualify as a successful unsupported crossing. Thus the argument does not move from a failed practical purpose to an absence of objecthood. Every established determinate physical incidence necessarily incurs the work; what varies between assessments is the substantive relation and whether the proposed use qualifies under its original conditions.

An actual record supplies a further instance. Its fitness to record an occurrence e depends on its actual relation to e ; keeping a label while substituting a different occurrence does not retain that relation. The record itself can remain a determinate physical object even when an evidential claim made with it is mistaken. Discovering and reporting its qualification are further acts, answerable to the physical conditions. The four roles are therefore not made conditional on anyone choosing an assessment convention or adopting the kernel terminology.

The objection nevertheless identifies the real constitutive burden. Numerical identity alone does not supply every physical law or substantive use-condition. The domain supplies those conditions. The argument is that determinate incidence under them necessarily differentiates its actual relata, its qualification for the relation, the conditions on their constructional combination, and its original occurrence. Removing one of these distinctions loses corresponding work in identifying what actually occurs or what a proposed use is entitled to carry forward. The reconstruction below makes that claim explicit. Its justification must be assessed at these steps; introducing a predicate named “adequate” would not by itself establish it.

This also limits (A2)’s quantification to evaluated assessments under complete stated conditions. It does not require every proposed question to have a unique result or a terminating decision procedure. An incompletely specified experiment may require an outcome set rather than a selected outcome; an undecided search may supply no assessment. Such cases do not license a discordant answer to an already determinate question. Result determinacy for the question actually answered remains a separate, explicit premise of Section 6.

3. Necessity, Closure, and Functional Minimality

Proposition 2.4 has defended the passage from actual determination to (A1)–(A4). The following theorem identifies the functions required to realize those commitments. It makes the second necessity step explicit and retains the first as its stated ground:

$$K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}.$$

Reference fixes and preserves what is concerned. Standing is the performance’s eligibility for a specified warranted use. Irreversibility fixes the original historical assessment against replacement by later repair. Admissibility controls whether the proposed reliance is warranted with these requirements jointly respected. These are functional positions, not four primitive axioms, four pieces of hardware, or four independently stored fields.

THEOREM 3.1 (Necessity of the four-role kernel). *Every target-adequate realization supplies reference, standing, irreversibility, and admissibility. Consequently every non-degenerate determinate constructional object or incidence necessarily instantiates all four roles. The realization may combine or distribute their implementation.*

PROOF. Fix an adequate realization. Its answer to the target question in (A1) yields a single-valued target assignment on the evaluated fragment, with the identification needed to preserve that answer under faithful redescription. This is the reference work Ref.

For an evaluated performance e and specified use u , let $L_D(e, u)$ express the favorable answer in (A2), with the stated conditions included in the query. The assessment is available as a basis for that use and remains attributable to e , its conditions, and u . Recording and preserving this eligibility is the standing work St. It need not license a different use or erase distinctions between performances that agree on this one answer.

Clause (A3) retains the bearer and original data of a historical claim. A later success can be attributed to a new performance or newly stated conditions, but cannot replace the data of the original claim without that replacement being acknowledged. Preserving this distinction is the irreversibility work Irr.

Finally, (A2) distinguishes warranted reliance from reliance unsupported by the assessed performance. Clauses (A1), (A3), and (A4) require this distinction to be enforced without target substitution, retrospective

repair, or transfer of an answer's warrant to a different bearer or question. This joint control of entry into constructional use is the admissibility work Adm. The mechanism recording standing may also perform this control; their functional distinction does not require separate machinery.

Each role has been recovered from an independently assessable requirement of the target. Proposition 2.4 supplies those requirements whenever non-degenerate determinate objecthood is established, completing the second assertion. $\square$

The argument does not obtain a rich physical or mathematical rule from four names. Its force is that any successful rule, however represented, must perform this work. A procedure that supplies the missing work under different terminology instantiates the role. An attempted omission that actually removes the work loses the associated target commitment. This exhausts functional omission without prejudging the mathematical content of any particular validator.

Example 3.2 (Derivation, correction, and delayed checking). Under implication elimination, the premises A and $A \rightarrow B$ can license B in a context in which the premises have standing. The inscription sequence $A, C \rightarrow B, B$ does not acquire that license merely because C is supplied later. The completed sequence is a different performance. By contrast, a later check of an unchanged proof object can establish that its original data already satisfied the rule. That is a change in knowledge or certification, not a repair of those original data. Both cases are governed by the kernel; the historical questions differ.

LEMMA 3.3 (Historical non-substitution). *Fix a realization, an original occurrence and its conditions, and a specified historical use. If that use is unwarranted on those data, supplying different data later does not make the unchanged original historical claim warranted.*

PROOF. The occurrence is identified without using its verdict by Definition 2.2. Clause (A2) supplies the answer to the fully specified original query. If the later intervention leaves that query and its data unchanged, the fixed realization supplies the same answer. If it changes the data, (A3) requires the change to be represented as such. Calling success on the changed input success on the unchanged original input would substitute a different bearer or conditions for the original claim. $\square$

This is irreversibility of the fixed historical attribution, not a prohibition on learning, retracting a mistaken report, revising a policy, or

authorizing a new use. A time-reversed physical evolution can itself be a further determinate process. It does not change which occurrence an earlier event claim concerned. No thermodynamic or dynamical irreversibility follows from the lemma.

PROPOSITION 3.4 (Role completeness). *For the declared constructional observation questions, every governance-relevant difference affects reference, standing, irreversibility, or admissibility, possibly several together. A further substantive condition can change an answer at one of these positions without constituting a fifth functional role.*

PROOF. A changed target affects reference. A changed eligibility for a specified use affects standing or admissibility. A substituted historical occurrence or original condition affects irreversibility and may also affect reference. A failure to preserve these answers, including their definedness, under a declared context affects their joint admissible use. These are precisely the kinds of questions in the independently declared frame. A difference affecting none of them performs no additional governance work for that frame. The cases overlap; no exclusivity is needed. $\square$

Completeness here is completeness for the stated target of construction. The observation family must therefore be justified by that target, not reduced after inspection of inconvenient examples. New evidence that a relevant question was omitted requires a richer frame and a renewed preservation claim. It does not become irrelevant by being absent from a chosen list.

DEFINITION 3.5 (Coverage and functional deletion). A package covers a role when its mechanisms jointly perform that role's work. A functional deletion removes the work from the whole package, including implicit, derived, or distributed implementations. Removing a name, while another mechanism still performs its work, is not a functional deletion.

THEOREM 3.6 (Functional minimality). *No package obtained by functionally deleting any kernel role remains sufficient for non-degenerate determinate construction.*

PROOF. If the remaining package were sufficient, Theorem 3.1 would reconstruct all four roles from its realization, including the supposedly deleted work. Theorem 2.8 and Corollary 2.9 make the alternatives exact for the independently specified original observation work. Either the remaining mechanisms still recover the original profile, in which case the

relevant function has not been deleted, or an original required distinction is merged, a true answer omitted, or a false answer supplied. Each such witness defeats the claimed realization. Thus genuine removal of the work and sufficiency for that same work cannot coexist. No claim that four physically separate components are required enters the argument. $\square$

The bridge example in Section 1.1 makes the losses concrete. Without reference, a different test can answer for the original one; without standing, appearance can substitute for actual qualification; without admissibility, outside support can be counted as the bridge's own performance; without irreversibility, repair can replace the original failed trial. A role has not been removed if another mechanism still performs its work. These losses can overlap because the four roles govern one construction together.

PROPOSITION 3.7 (Mutual closure of the realized roles). *In a complete constructional realization, admissibility, standing, reference, and irreversibility form one closed interface: their intended work cannot be retained while denying the work to which it is accountable.*

PROOF. Admission of a specified use must confer eligibility for that use, or the favorable assessment is unavailable as a constructional basis. Such eligibility must concern a determinate performance and target, or nothing determinate is eligible. Its target-bearing historical attribution must remain about the original occurrence and conditions, or a later replacement can change what the eligibility concerned. Preserving this attribution requires distinguishing faithful use from substitution, bringing the analysis back to admissibility. Thus each passage preserves the work of the complete interface. $\square$

This closure describes the constructional senses of the roles. Bare denotation need not confer standing; physical non-rewindability need not assess a use. Conversely, one sufficiently rich clause may implement the whole interface. Functional minimality is compatible with such compression because it concerns indispensable work, not logical independence of four axioms. A derivation of one implementation component from others can be a genuine mathematical achievement while leaving its role fully instantiated.

4. The Authorization Boundary of a Proposed Lower Generator

A representation of a rule and authority to use that representation are different achievements. Computing a predicate can explain how a validator operates. A soundness theorem can justify applying it. The question for a purported governance-free generator is where its output first acquires warrant for the original target-domain use. The following finite argument isolates that point without asserting that mathematical explanation or implementation is impossible.

DEFINITION 4.1 (Support certificate and semantic authorization cut). A support certificate is a finite acyclic dependency graph of premises and rule applications, with their interpretations stated. A node can describe data, compute a value, or assert a warranted target-domain use. A target-authority node is one whose interpreted conclusion supplies warrant for a specified use on D . A semantic authorization cut, called an authorization cut below, consists of the first such nodes along the supporting dependency paths. At a cut node, authority is either an already warranted premise or the conclusion of a rule whose application licenses a target-domain use. The cut is defined pathwise; a first authority node on one path may have an authoritative ancestor along another incoming path.

Here “cut” names a boundary of authority in an interpreted dependency graph. It is not Gentzen’s cut rule: a proof with no applications of that syntactic rule can still contain warranted premises and inferences introducing target-domain warrant. Eliminating intermediate lemmas can reorganize a proof without removing this semantic work. The claim below is about where warrant enters a finite acyclic certificate, not about failure of proof-theoretic cut elimination.

The distinction concerns interpreted support, not vocabulary. A rule can authorize a use without containing the word “admissible.” Conversely, printing a sentence that says a use is authorized does not establish its truth or warrant. A certificate is governance-free in the relevant sense only if it has neither an already warranted target-use premise nor a rule that introduces target-use authority. Merely hiding that rule in a decoder, soundness claim, trusted oracle, or translation does not remove it.

LEMMA 4.2 (Finite semantic authorization-cut lemma). *Every finite certificate warranting a target-domain use has an authorized premise or a rule instance introducing target-domain authority. A certificate having neither cannot warrant that use.*

PROOF. Restrict the dependency graph to the ancestors of the warranted conclusion. This finite acyclic graph contains at least one target-authority node, namely the conclusion. Choose a target-authority node with no target-authority ancestor. If it is a premise, the first alternative holds. Otherwise it results from a rule application with no target-authority premise; that application introduces target-domain authority, giving the second alternative. Independently, taking the first authority node on each source-to-conclusion path yields the pathwise authorization cut. Those pathwise first nodes need not all be globally minimal authority nodes; the existence argument requires only the globally minimal node just chosen. A graph with neither kind of node cannot contain the warranted conclusion. $\square$

The lemma is an elementary dependency fact. Its relevance comes from requiring the warrant to be identified rather than conflated with a computed string. A rule that introduces authority can be justified by mathematics or by the independently specified conditions of the domain. The lemma does not say that such a justification must be circular. It says that, wherever justification succeeds, the claimed target-use authority is supplied somewhere in the complete account. Identifying that point directs attention to the premise or rule whose warrant must be examined; marking a node as authoritative does not itself justify its interpretation. Finiteness and acyclicity support the existence of a first authority node. Infinitary or cyclic support requires its own analysis and is not classified by this lemma alone.

THEOREM 4.3 (No governance-free faithful lower generator by finite certificate). *No finite certificate generates authority for the original fixed-domain uses while remaining governance-free. Every successful proposed generator represented by such a certificate has an authorization cut at which the relevant kernel work is instantiated. A faithful external encoding, computation, or metatheory does not eliminate that requirement.*

PROOF. By Lemma 4.2, a successful certificate contains an authorized premise or an authority-introducing rule. In the first case its support already includes target-domain warrant. In the second case the introduction of authority is part of the generator's complete interface. To make the connection explicit, interpret a node by its original subject x_n and its warranted use by the original witness relation $J(x_n, u, a)$. The minimal authority node supplied by the lemma then has an actual

witness $J(x_c, u_c, a_c)$. Proposition 2.6 constructs its positive qualification; Lemma 2.7 fixes the permissible ground of reliance; and Theorem 2.8 determines what any encoding of that original work must preserve. No field asserting that the cut already possesses the kernel enters this argument. For that interface to warrant the original use, it must concern the original target and performance, confer the specified eligibility, retain the original historical data, and distinguish warranted reliance from substitution or unsupported reliance. Theorem 3.1 identifies this as kernel work. Neither case is governance-free.

An external system adds a translation to the account. If its interpreted output warrants the original use, the translation and its justification participate in the same authorization cut. If it does not, the output has not yet supplied the claimed target-domain warrant. The argument is independent of the syntax or location of the computation. $\square$

This is structural non-derivability as authority bootstrap: the required governance work cannot disappear from a successful justification of its own target uses. It is not syntactic unprovability of sentences containing kernel terms, a ban on simpler definitions, or a claim that an implementation existed chronologically before being built. A new checker may first instantiate the relevant work when it begins operating. An invariant may provide a substantive explanation of an admission criterion. In each case its successful application retains the authorization interface. The result concerns that interface, not the impossibility of every lower-level causal or mathematical account.

PROPOSITION 4.4 (Bare data underdetermine a policy). *If two adequate realizations agree on a specified reduct of their data and disagree on a target-use answer, no function of that reduct alone recovers the correct answer for both.*

PROOF. Such a function receives the same input in both realizations and must return the same output. It therefore cannot equal both differing answers. For a finite instance, take immutable occurrence records z, p, n , one target and use, and identity contexts. One realization warrants the use for z, p ; another only for z . Both reject n , preserve their historical answers, and satisfy (A1)–(A4). Their trace records coincide; their governing answer for p differs. $\square$

This example concerns a base frame on which the substantive use condition has not yet been fixed. It does not assert that two different

answers are both correct under one complete contract. The distinction establishes the precise reach of the necessity theorem: both realizations perform the four roles, while the additional question which realization fits a given domain remains substantive.

5. One Fixed Contract and Exact Governance Equivalence

The expression “same domain” has two uses that must remain distinct. The same base frame supplies common candidates and questions; different answers can be compared on it. The same complete governed contract additionally fixes the substantive answer conditions and their contextual behavior. The resulting profile records their determinate answers; it is not an independent authority that supplies their justification. Universal role necessity concerns all adequate realizations. Exact governance equivalence concerns realizations faithful to one such contract.

DEFINITION 5.1 (Complete governed contract). A complete governed contract specifies the base observation frame and independently warranted answer conditions for all its relevant questions, including partial-operation definedness and the contexts required for constructional use. The following profile records the resulting determinate answers. If Y_q is the ordinary answer set for question q , adjoin a fresh undefined value $\perp$. Its observation profile is

$$\Pi_D : E_D \longrightarrow \prod_{q \in \mathcal{Q}_D} (Y_q \sqcup \{\perp\}).$$

Completeness is relative to the declared construction problem: every distinction that can affect its target, use, historical attribution, or stipulated contextual behavior must be represented. Governance equivalence means preservation and reflection of this complete profile under the proposed identification.

Here $\perp$ records the declared observation’s undefinedness; interpreting it as a settled refusal additionally requires the complete-outcome evidence of Section 6.

The contract records substantive conditions; it does not manufacture their justification. Mathematical assumptions, empirical identifications, or domain-specific validation rules must be exposed when they determine its answers. Equality of one present assessment cannot substitute for

equality of the full profile. In particular, two performances can both be eligible for one use while differing in eligibility for another.

PROPOSITION 5.2 (Exact preservation and the limit of role uniqueness). *Packages exact for the same complete governed contract agree on every declared governance answer after their representations are identified. Coverage of the four roles alone does not entail this agreement. A condition that changes a declared answer changes the contract; a condition that preserves every answer can refine implementation without changing the contract’s governance content.*

PROOF. At the level of the recorded contract, exactness requires each package to preserve and reflect every coordinate of Π_D , including undefinedness. Their answers consequently agree through the common contract. Role coverage identifies the kinds of work a package performs, but Proposition 4.4 exhibits covered packages with different answers on a common base frame. Finally, changing an answer changes a coordinate of the contract, whereas preserving all coordinates leaves its governance content unchanged. $\square$

The canonical quotient theorem (Theorem B.3) makes the representational consequence precise: exact reduced realizations of one complete profile are uniquely isomorphic over their common inputs. Its congruence conditions also state when partial contexts descend to the quotient. Those are mathematical consequences of the specified observation structure. They do not provide the conceptual premise of objecthood, and they do not derive a unique substantive policy from role coverage alone. Section 6 provides the prior justification for agreement: individual answers incident under independently fixed determinate rules agree at the same bearer and question. Thus the contract is a representation of settled governance, not a proof-generating assumption that all encodings are already faithful. Together the levels yield the intended architecture: determinate objecthood necessarily incurs the kernel; independently warranted answer conditions determine its fixed-domain realization; exact quotients represent the resulting effects.

6. Conservation of Governed Representations

The kernel governs an actual determinate object and its specified uses. The original bearer and complete question are fixed before representations

are compared. A genuine alteration of their identity-bearing data presents a different construction or question. An inscription that continues to claim the original instead has to establish its answer-standing there; changing that inscription does not, by itself, change its subject. This section proves the corresponding conservation theorem without presupposing equality of observational profiles. The argument fixes the bearer and question independently, establishes each proposed answer through its own incidence evidence, and derives equality from determinacy of the relevant result. It applies to every admissible presentation satisfying those conditions, irrespective of its encoding or implementation.

6.1. The result incidence and its original grounds

DEFINITION 6.1 (Governed result incidence). Let B be a type of bearers, Q a type of complete governed queries, and V_q the result type of query q . A result incidence is an independently specified relation

$$J(b, q, v), \quad b \in B, \quad q \in Q, \quad v \in V_q.$$

The bearer includes the original occurrence and identity data relevant to the task. The query includes its scope, requested role, original conditions, and substantive parameters. These are fixed before the proposed result is assessed. The relation is determinate on the evaluated domain when

$$J(b, q, v) \wedge J(b, q, w) \implies v = w. \quad (1)$$

A result has standing as an answer at (b, q) precisely through its warranted incidence there, with that incidence established from the original domain rules.

The relation is not defined by comparing representations or by selecting whichever outputs agree. It may be specified by derivation rules, an evaluation relation, or independently fixed conditions of a physical incidence. Its determinacy must concern this result relation. Unique reference to the input object alone does not prove that every question one might ask about it has a unique result. A relation admitting several possible outcomes requires further substantive specification before it supports the determinate-result theorem. Adding the proposed answer to the query solely to force uniqueness would not discharge that requirement.

Representations may belong to any type R . On the fragment whose identification has been established, write $\beta : R \rightarrow B$ for their bearer

assignment. An original reader produces $F_r(q) \in V_q$. The representations may use different algorithms; their algorithms and inscriptions can be included in r . Neither equality of outputs nor an equation factoring F through β is part of these data. A claim that $F_r(q)$ has standing must instead supply its own warrant for

$$J(\beta(r), q, F_r(q)).$$

This is correctness of one answer under the original rules. It is distinct from a comparison with another reader's answer.

Theorem 3.1 already establishes kernel governance for the determinate objecthood at issue. No further permission determines whether it applies here. The present proof makes explicit what preserving the object and using an answer with standing require. It does not use an assumed all-observations preservation equation to obtain the conservation equation below.

6.2. Local conservation and the complete query family

THEOREM 6.2 (Same-bearer, same-query conservation). *Suppose J is determinate. Let two original representation uses retain the same bearer b and complete query q , with outputs $v, w \in V_q$. If each output has established incidence at that bearer and query, then $v = w$.*

PROOF. The identification of the bearer and query places the two independently established incidences in the same fibre: $J(b, q, v)$ and $J(b, q, w)$. Apply (1). If the presentations use equal query indices written differently, first transport their results along that equality into the same result type. No comparison between outputs was needed to establish their common bearer or query. $\square$

The theorem is uniform over readers and presentations. It does not require a list of permitted encodings or a grammar generating every observation. Its premises must nevertheless be true of the uses being compared. The bare fact that two inscriptions exist establishes neither their common bearer nor the standing of their proposed answers in the claimed role.

THEOREM 6.3 (All-query representation conservation). *Let J be determinate for every query in an independently fixed family Q . Suppose $\beta(r) = \beta(r')$, and the two original profiles are individually sound:*

$$\forall q \in Q \quad J(\beta(r), q, F_r(q)), \quad \forall q \in Q \quad J(\beta(r'), q, F_{r'}(q)).$$

Then $F_r = F_{r'}$ as elements of $\prod_{q \in Q} V_q$. Consequently, if every member of an established class of representations supplies such a sound original profile, the equality holds for every pair in that class with the same bearer.

PROOF. Fix any q . The bearer equality and the two incidences satisfy Theorem 6.2, so $F_r(q) = F_{r'}(q)$. Since q was arbitrary, function extensionality gives equality of the dependent profiles. Apply this argument to each pair for the final assertion. $\square$

The family Q can be infinite and need not be generated by finite syntax. Its completeness is justified by the original task: all relevant uses, reports, diagnostics, and application questions must remain represented. This universality concerns warranted answers, not the ability of a finite computation to discover every answer. If a reader has established answers only on a fragment, the local theorem applies there; full profile equality requires the stated soundness throughout the family. Writing that soundness obligation as a certificate field does not establish it. Its warrant must come from the original construction or a sound verification of its individual rules.

COROLLARY 6.4 (Discordance excludes joint standing). *At a fixed determinate bearer and query, two different proposed results cannot both have incidence. If one result v is known to be incident, every proposed $w \neq v$ lacks incidence as an answer at that same bearer and query.*

PROOF. Joint incidence would imply $v = w$ by Theorem 6.2, contradicting discordance. With incidence of v fixed, assuming incidence of w supplies exactly that contradiction. $\square$

This conclusion is role-specific. It does not deny that the discordant inscription is a determinate object in its own right, or that it can have standing for storage, diagnosis, or another use. What it cannot retain is standing as the discordant answer to the unchanged query about the unchanged bearer. Nor does failure of a presentation remove standing from the original object. A different determinate construction remains

necessarily governed by the kernel, while its different bearer or query prevents importing its answer into the unchanged original assessment. A proposed new classifier that changes a governed answer must therefore expose a changed bearer or query, or fail the standing claim for at least one answer. If it changes none of the answers, it supplies no additional distinction in their governed profile.

The bridge under changed conditions. In Section 1.1, supporting the bridge with a finger changes the conditions of the crossing. Its successful trial can have a warranted positive answer while the unsupported trial has a warranted negative one: the complete queries differ. A report that conceals the finger and claims success for that same unsupported trial has not supplied a second authority at the original locus. Once the unsupported outcome is independently fixed, Corollary 6.4 excludes standing for the discordant report as an answer there. An observer’s uncertainty about whether support was used is a further epistemic question; it is not a third outcome of the fully specified original trial.

6.3. Completed outcomes and computational absence

Partial evaluation requires a distinction between a settled negative result and an unfinished attempt to obtain a result. Let $J_0(b, q, v)$ be the incidence of a base result, with $v \in W_q$. Define a complete outcome relation on $\text{Option}(W_q)$ by

$$\begin{aligned}\hat{J}(b, q, \text{some}(v)) &\iff J_0(b, q, v), \\ \hat{J}(b, q, \text{none}) &\iff \forall v \in W_q \neg J_0(b, q, v).\end{aligned}$$

The meaning of none is absence of every base result, not absence of a returned message.

PROPOSITION 6.5 (Determinacy of complete outcomes). *If J_0 is determinate, then $\hat{J}$ is determinate. Accordingly two standing-bearing complete outcomes at one bearer and query agree in definedness and, when present, in their base value.*

PROOF. Two present outcomes agree by determinacy of J_0 . Two absent outcomes are the same constructor. A present outcome and an absent outcome cannot both be incident: the latter denies the base incidence supplied by the former. These cases exhaust the optional result type. Apply Theorem 6.2 to $\hat{J}$. $\square$

A timeout, pending check, or lack of evidence cannot be substituted for the universal negative in the second clause. If a known base result exists, such a computational absence has no standing as the complete semantic outcome none. Classical reasoning can establish that a base result exists or none does without supplying an effective procedure to decide which. Conversely, an independently verified partial-reference evaluator can compute the outcome when its domain permits that computation.

The result type must retain the original distinctions. If a rejection reason is observed, specify it as part of the result or as another query. A negative answer with a diagnostic can itself be an incident, standing-bearing report. Its standing as a report does not grant the rejected construction standing for the refused use. These roles must remain distinct when complete outcomes are transported.

6.4. The actual canonical status

For a fixed locus $\ell = (b, q)$ of a determinate incidence J , let

$$\text{Ref}_\ell = \{v \in V_q \mid J(b, q, v)\}, \quad \text{Fail}_\ell \not\Longleftrightarrow \forall v \in V_q \neg J(b, q, v).$$

A reference retains both its value and its incidence evidence. Define the canonical status type St_ℓ by constructors $\text{standing}(r)$ for $r \in \text{Ref}_\ell$ and $\text{failure}(h)$ for evidence h of Fail_ℓ . Proposition proofs are identified proof-irrelevantly, as in the Lean formalization.

Proof irrelevance here identifies evidence of an already fixed proposition; it does not identify distinct proof trees, derivation occurrences, or diagnostic records. Those objects remain in the bearer and result data whenever the original question distinguishes them. Equality of canonical status therefore does not settle the general problem of identity of proofs discussed in Section 1.

THEOREM 6.6 (Unique canonical status at a fixed locus). *The canonical status type St_ℓ is a subsingleton: any two of its inhabitants are equal.*

PROOF. Two standing statuses have equal reference values by determinacy and equal incidence evidence by proof irrelevance. Two failure statuses have equal failure evidence by proof irrelevance. A standing status and a failure status cannot coexist because the former supplies an incidence denied by the latter. Thus every possible pair is equal. $\square$

COROLLARY 6.7 (Every canonical consumer preserves its result). *With additional arguments fixed, every total, partial, or dependent consumer of canonical status agrees on any two inputs at the same determinate locus.*

PROOF. Theorem 6.6 gives $e : s = t$. For a total consumer $G : \text{St}_\ell \rightarrow Y$, congruence gives $G(s) = G(t)$. Taking $Y = \text{Option}(Z)$ also preserves partial definedness. For a dependent consumer $G : \prod_s Y(s)$, equality elimination gives

$$\text{transport}_Y(e, G(s)) = G(t).$$

The same argument applies to dependent optional results. A proposition-valued application domain is preserved along e , and proof irrelevance identifies its evidence when comparing a domain-indexed result. Finite tuples of canonical arguments agree componentwise. $\square$

An actual reference or failure supplies an inhabitant; a partial-reference presentation supplies one by evaluating its reference function. Theorem 6.6 alone asserts at most one inhabitant. Positive and negative polarity tags can both occur across different loci with the corresponding witnesses, while one fixed locus has only its actual status. An incident negative report, viewed at the report’s own result-bearing locus, has positive standing there. Constructor polarity must not be confused with the content of that report.

The canonical theorem is stronger than an assumption that selected consumers preserve a bit: it covers every function on the actual semantic type. A raw encoding type E_ℓ with a decoder into St_ℓ is different. Equal decoded values do not make its encodings equal, and a raw consumer may inspect an erased field. To apply conservation, establish that the original consumer’s result has incidence for the unchanged bearer and query; merely possessing a decoder is insufficient.

6.5. Worked example: an independently deterministic proof checker

Consider formulas built from a decidable alphabet of atoms and implication. Fix an ordered finite assumption list Γ and finite annotated proof trees generated by h_i and $\text{mp}(d_1, d_2)$. The first cites assumption position i ; the second invokes implication elimination. A bearer records the original occurrence, Γ , and the tree d , before checking succeeds or

fails. Define the result incidence independently of encodings by these rules:

$$\frac{\Gamma[i] = A}{\Gamma \vdash h_i \Downarrow A}, \quad \frac{\Gamma \vdash d_1 \Downarrow A \rightarrow B \quad \Gamma \vdash d_2 \Downarrow A}{\Gamma \vdash \text{mp}(d_1, d_2) \Downarrow B}.$$

The query is inference under these fixed rules and assumptions; its result type is the formula type. No equation comparing two implementations occurs in these rules.

PROPOSITION 6.8 (Determinate checking and complete outcome). *For each fixed tree and assumption list, the inference relation has at most one result. A structural checker returning $\text{Option}(\text{Formula})$ computes its complete outcome.*

PROOF. Induct on the tree. At h_i , lookup in one fixed list has at most one value. At an implication-elimination node, apply the induction hypothesis to the left subtree in any two putative derivations. Its conclusions $A \rightarrow B$ and $A' \rightarrow B'$ are equal, so the uniquely parsed implication constructor gives $A = A'$ and $B = B'$. The right-subtree derivations must also agree by induction. The parent conclusions are consequently equal.

Define the checker recursively. It returns the cited formula for an in-range assumption and none otherwise. At an implication-elimination node, it checks both children. It returns B exactly when the left result is $A \rightarrow B$ and the right result is syntactically A ; otherwise it returns none. Structural recursion terminates. Induction proves that a returned formula has a derivation and that every derivation produces its stated formula. If a child returns no formula, the induction hypothesis excludes the premise required by the corresponding rule. An out-of-range citation, non-implication left result, or mismatched antecedent likewise admits no rule instance. Thus the absent outcome certifies absence of every derivable result. $\square$

Now take two original encodings, such as a prefix serialization and a parenthesized tree serialization. A parsing proof can establish independently that each records this same Γ and d , with the same original occurrence identification. Running the verified checker after either parser establishes each output's complete-outcome incidence. Proposition 6.8 and Theorem 6.2 then derive equality of the outputs. The proof did not assume that the parsers or observers agreed on their answers; it checked their source identifications and each evaluation separately.

A direct checker for either serialization can also be used. Its positive answers require soundness against these inference rules; agreement of complete optional outcomes additionally requires that a negative answer establish absence of every derivable result. A sound but incomplete search can leave a case unresolved without producing a settled negative answer. An implementation reporting success for a mismatched antecedent has failed its soundness obligation. Its production of an inscription remains a separate event; it does not supply a new valid answer for the original proof tree. If the reporting task observes additional diagnostic information, that information needs its own fixed result specification. The simple absence result above does not silently preserve distinctions the checker never specified.

6.6. Universal scope and exclusion of an additional authority

THEOREM 6.9 (Governed representation conservation). *For every independently specified determinate result incidence, every pair of actual admissible uses of the same bearer in the same complete query role has the same result. This holds jointly for the complete original query family whenever the profiles supply standing-bearing answers throughout that family. A discordant proposed answer cannot retain both the original bearer/query and standing as an answer there. The claim is independent of the number, syntax, or implementation of the representations.*

PROOF. Actual standing-bearing answers provide their respective incidences. Theorem 6.2 gives local equality; Theorem 6.3 gives the complete profile statement. Corollary 6.4 excludes joint standing of a discordant pair. None of these arguments restricts the representation type or enumerates the query family. $\square$

The result concerns the intended same-object uses, not arbitrary coding rules asserted to describe them. Kernel governance is necessary for those determinate uses; identifying their bearer, fixing their question, and establishing their incidence expose that necessity in the proof. A purported third same-object authority either leaves every governed answer unchanged, alters the object or question, or supplies a discordant answer without the claimed standing. This excludes the alleged additional authority without identifying distinct objects, suppressing diagnostics, or treating every positive label as the same construction.

7. Fixed-Domain Exhaustion and Proof Dependencies

THEOREM 7.1 (Kernel necessity and same-object exhaustion). *Every non-degenerate determinate constructional object or incidence instantiates Admissibility, Standing, Reference, and Irreversibility as functional necessities. For its independently specified determinate constructional assessments:*

- (i) *any standing-bearing answers yielded by actual admissible presentations agree at the same bearer and complete query;*
- (ii) *a proposed additional classification that changes such an answer either changes the bearer or question, or lacks the claimed answer-standing at the unchanged assessment; a classification changing no answer adds no distinction to the governed profile;*
- (iii) *a later performance cannot repair an original negative historical answer while preserving that occurrence and all of its original assessment conditions;*
- (iv) *a realization preserving and reflecting the complete original observation profile has the same fixed-domain governance content, regardless of the number or names of its implementation components. No functional deletion of a kernel role retains the original complete constructional job.*

These assertions hold uniformly over representations and query families; they do not require the latter to be generated by a finite observation language.

PROOF. Theorem 3.1 reconstructs the four roles from the commitments of non-degenerate determinate objecthood. For (i), an actual standing-bearing answer supplies its incidence under the original rules, and Theorems 6.2 and 6.3 give the local and full-family conservation statements. For (ii), a difference either occurs at a changed bearer or query, or at the same one. In the latter case Corollary 6.4 excludes standing of the discordant answer alongside the established answer. If no answer changes, the governed profile is unchanged by definition of that profile; this last branch asserts no new authority. Clause (iii) is Lemma 3.3, with the occurrence identified before its verdict. Exact agreement for (iv) follows from the complete profile and Proposition 5.2; functional deletion is excluded by Theorem 3.6. The only finiteness used elsewhere in the paper

concerns particular certificate, syntax, or dependency constructions, not the semantic quantifiers in (i)–(ii). $\square$

The theorem concerns a complete constructional job. A sound checker that has established results only on a fragment can retain standing for those results without deciding every query. It has not thereby supplied an exact complete realization. Conversely, a system claiming complete assessment must justify both its positive results and its settled negative answers. An unexamined case is neither an additional standing value nor a demonstrated semantic refusal.

7.1. The direction of support

Proposition 2.4 defends the passage from actual determinate realization to (A1)–(A4). Results 2.6–2.9 construct its semantic qualification, identity-transfer, exact-realization and failure results; Theorem 3.1 reconstructs the four functions from those commitments. These necessity steps precede the representation analysis. The conservation theorem derives equality from independent answer incidence and result determinacy; it does not derive either premise from the equality it proves. The exact quotient then records which information is preserved, and the primitive theorem gives an implementation method. This order avoids borrowing foundational authority from downstream bivalence or occupancy results that already presuppose it.

Claim	Grounds of the result
Necessary kernel governance	Original incidence and identity; constructed qualification, the identity-transfer lemma, the exact-realization criterion, and reconstruction of the four functions.
Functional minimality	Original-profile recoverability and witnessed omission, surplus, or fibre collision; implicit and distributed substitutes count as retained work.
Finite authorization boundary	Finite acyclic interpreted support certificates; the authorization-cut lemma and reconstruction of the required kernel work.
Same-object conservation	Independently fixed bearer and complete query; determinacy of the relevant result incidence; each answer's own warrant.
Complete-outcome preservation	Present outcomes are incident results; an absent outcome establishes absence of every base result.
Binary admission	A specified evaluated use; both represented polarities require positive and negative witnesses.
Binary contextual reduction	Retained bearer/query identification and original-outcome incidence, or separately verified primitive descent with original-observation coverage.
Exact semantic realization	The complete original observation family, with all standing parameters available in contexts.
Symmetry and retained support	The specified permutation action; or unchanged occurrences and required dependency edges, respectively.

These are different proof dependencies within necessarily governed construction. They are not optional permissions for the kernel to apply. In particular, the semantic realization theorem supplies a universal conservation argument for actual admissible presentations; it does not certify a data structure whose fields have no demonstrated relation to the object it claims to represent.

The remaining main-text analysis distinguishes binary admission from a complete profile and states the exact representation criterion. Appendices A and B supply the detailed implementation and structural proofs. Appendix C identifies the checked Lean results and their precise relation to the written argument. This separation makes the direction of dependence explicit while retaining all supporting results in the reference edition.

8. Semantic Status and Faithful Binary Representation

Kernel governance is necessary for non-degenerate determinate construction. The semantic conservation theorem supplies its representation-independent consequence: incident answers to the same complete question about the same bearer agree, for every admissible presentation. This universal statement neither enumerates encodings nor restricts observations to a finite syntax. A proposed raw-code reduction realizes it when the reduction retains the independently identified bearer and query and the original outputs have standing as answers to that query. We distinguish this semantic result, the binary admission verdict, and the representation theorem connecting them.

8.1. Complete evaluation and actual semantic status

DEFINITION 8.1 (Complete evaluation locus). An evaluation locus fixes the act and candidate identity, target, requested operation and role, input objects, substantive parameters, original conditions, provenance, and relevant boundary, continuation, and failure conditions. For the evaluated loci $\mathcal{E}_D$, let G_D contain those licensed for that specified use. Their admission indicator is

$$a_D : \mathcal{E}_D \rightarrow \{0, 1\}, \quad a_D(e) = 1 \iff e \in G_D.$$

An unfinished computation is not an evaluated negative answer merely because its implementation has returned no result.

PROPOSITION 8.2 (Binary admission and a unique actual status). *The admission distinction has the unique characteristic representation a_D . Its represented verdicts number at most two, and exactly two precisely when both licensed and unlicensed loci occur. At one fixed determinate locus, the canonical semantic status is unique; every consumer of that actual status preserves its observations, including partial domains.*

PROOF. Membership in G_D and its complement give the characteristic map and its image. For canonical status, Theorem 6.6 and Corollary 6.7 apply: standing carries an incident reference, failure carries absence of incidence, determinacy identifies positive references, and the two alternatives cannot coexist at one locus. $\square$

Thus a completed locus has one actual status, not two simultaneously inhabited status classes. Classical existence of that status does not

supply a computable evaluator. Across loci both admission outcomes may occur, while the loci retain their distinct substantive content. Candidate admission, role eligibility, compatible occupation, and realization are different questions; a positive answer to one does not answer the others.

8.2. Typed representations and their original observations

Retain an independently specified substantive base B . For $b \in B$, let V_b be the raw status presentations to be compared and let $\epsilon_b : V_b \rightarrow \{0, 1\}$ record admission polarity. Write $S = \coprod_b V_b$. The base retains the original object, act, role, input, and scope data. It is not manufactured by moving an erased code or the desired full observation profile into a new field. Equal polarity alone does not identify bearers or complete queries.

DEFINITION 8.3 (Original status observations). Fix the original licensed observation family $\mathcal{C}_b$, including admission, use, standing, commitment, reportability, and failure observations at issue. Totalize each partial observation $O_c : V_b \multimap Y_c$ by its definedness symbol $\perp$. Put

$$\Phi_b(s) = (O_c^\perp(s))_{c \in \mathcal{C}_b}, \quad s \approx_b t \iff \forall c \in \mathcal{C}_b \ O_c^\perp(s) = O_c^\perp(t).$$

THEOREM 8.4 (Canonical profile realization). *There is a unique bijection over the profile map $V_b / \approx_b \cong \text{im}(\Phi_b)$, sending $[s]$ to $\Phi_b(s)$. Every original observation is realized by its coordinate projection, with its original definedness.*

PROOF. By definition, equivalence is equality of profiles. The displayed map is therefore well-defined, injective, and surjective onto the image. Its value on each class is forced, and projection recovers the original observation. $\square$

THEOREM 8.5 (Exact binary factorization). *The following are equivalent:*

- (i) $s \approx_b t$ if and only if $\epsilon_b(s) = \epsilon_b(t)$;
- (ii) every original totalized observation is constant on polarity fibres;
- (iii) an injective map $H_b : \text{im}(\epsilon_b) \rightarrow \text{im}(\Phi_b)$ satisfies $\Phi_b = H_b \circ \epsilon_b$, with ϵ_b corestricted to its image.

Under these conditions the contextual quotient is canonically $\text{im}(\epsilon_b)$.

PROOF. Condition (i) gives (ii). Under (ii), collect the observations of any representative of a represented polarity to define H_b . Constancy makes this well-defined. The admission coordinate distinguishes different

polarities, giving injectivity. Condition (iii) equates profiles exactly when polarities agree, yielding (i). $\square$

The semantic conservation theorem provides a direct way to establish these preservation claims for actual presentations. The criterion remains useful for specifying precisely what a proposed representation erases.

8.3. Connecting faithful representations to semantic conservation

Use the original many-sorted partial interface. Ordinary sorts retain acts, targets, roles, other inputs, and observed outputs. Reduce the status sort by

$$\pi_S(b, s) = (b, \epsilon_b(s)), \quad \bar{S} = \{(b, i) : i \in \text{im}(\epsilon_b)\},$$

and use identity reduction on ordinary sorts. All reduction maps π_A are surjective onto their reduced sorts.

DEFINITION 8.6 (Primitive descent). A primitive $f : A_1 \times \cdots \times A_k \rightarrow A$ descends when equal reduced input tuples imply

$$f(\vec{x}) \downarrow \iff f(\vec{y}) \downarrow, \tag{2}$$

$$f(\vec{x}) \downarrow \implies \pi_A(f(\vec{x})) = \pi_A(f(\vec{y})). \tag{3}$$

The inventory includes applicability, branching, standing, reuse, reporting, and failure tests, not just the admission projection.

THEOREM 8.7 (Semantic discharge of primitive descent). *For each original primitive, independently specify its bearer map β_f , complete-query map κ_f , and a determinate result-incidence relation J_f on those bearers, queries, and complete outcomes in $\text{Option}(\bar{A})$. Suppose:*

- (i) *equal reduced input tuples identify the same β_f and κ_f from the original identity and query data;*
- (ii) *every compared original application has a justified complete outcome: $\text{map}(\pi_A, f(\vec{x}))$ is incident at $(\beta_f(\vec{x}), \kappa_f(\vec{x}))$.*

Then every such primitive satisfies both descent conditions collectively.

PROOF. Take equal reduced inputs. Clause (i) identifies their bearer and query; clause (ii) supplies the two original outcome-incidence witnesses. Theorem 6.2 therefore identifies the complete optional outcomes. Equality of optional outcomes preserves whether they are absent or present, proving (2); equality in the present case proves (3). $\square$

The premises are statements about the independently fixed semantics and each original answer, not equations asserting agreement between codes. They must be established from original rules. Neither the bearer/query maps nor the result relation may be chosen retrospectively to encode the desired agreement; in particular, erased raw labels are not extra identity data. Determinacy of reference to an input object alone does not establish determinacy of every proposed result relation. Likewise, determinacy does not validate an arbitrary reader. For an absent result, Proposition 6.5 requires absence of every licensed base result; a timeout or pending computation does not satisfy that obligation merely by returning `none`.

This is a conditional realization theorem with a universal semantic proof. Once the claimed identity and answer-standing facts are established, equality follows for every presentation, without a separate invariance postulate. The same argument applies directly to every original observation, including an unrestricted query family, through Theorem 6.3. No finite-language completeness premise is needed on that direct route. For the generic raw interface, the identity and output-incidence bridge must still be proved; calling a code neutral does not supply it.

8.4. Exact uniqueness and the retained construction

The profile quotient in Theorem 8.4 is exact: it identifies precisely the presentations that the original observations cannot distinguish. Every map constant on those classes factors uniquely through the quotient. Indeed, its value on $[s]$ must be its value on s , and class constancy makes that assignment well-defined. A surjective classification with exactly the same fibres is therefore uniquely isomorphic to the quotient through the classification maps. This is uniqueness of the exact realization of fixed governance content, not a claim that all domains have one substantive policy.

For constructional use, the original observations include application definedness and are closed under the relevant typed contexts, with standing parameters retained. Under those conditions the operations descend as well: replacing one argument at a time preserves its application domain and every observation of its result. The full partial-operation statement and proof are Lemma B.2 and Theorem B.3 in Appendix B. The quotient can have many classes. Only when the criterion in Theorem 8.5 holds does the specified status quotient reduce to represented admission polarities.

The implementation consequences are developed in Appendix A. Exact primitive descent gives a unique reduced partial algebra and is preserved by finite typed composition and ordinary Boolean conditionals. This supplies a constructive method for checking an interface. Direct semantic conservation remains available for an unrestricted original observation family. Neither route deletes observed diagnostic information: the reason for a failed trial, for example, remains part of the record wherever the original question observes it. Standing of a correct failure report is distinct from standing of the failed construction.

The further results in Appendix B concern licensed application domains, comparison paths, permutation symmetry, and retained dependency support. Each states its additional hypotheses. In particular, the symmetry result assumes the specified full permutation action, and the support results concern fixed required dependencies. These consequences develop the foundation; none is a premise used to establish the principal necessity and exhaustion theorem.

9. Interpretations and the Burden of a Counterexample

9.1. A failed construction and a later correction

Return to the bridge in Section 1.1. The loose joint gives way during the first trial. After repair, the car crosses in a second trial. The successful crossing belongs to the repaired arrangement and later occurrence; it does not change whether the original arrangement carried the car under its original support conditions. The fixed complete construction therefore distinguishes the original trial from its replacement. Calling an artifact the same bridge across that history does not identify its changed arrangement or later trial with the original complete performance. Both trials remain determinate physical events and necessarily incur the kernel's work.

Three queries distinguish repair from improved knowledge. The historical query asks whether the first trial satisfied its original conditions. The epistemic query asks what is presently known or correctly reported about that trial. The new-use query asks whether the repaired arrangement succeeds in its own trial. Repair can change the third answer; better evidence can change the second. Neither makes the original collapse a successful crossing while retaining the original occurrence and conditions.

Reference fixes the subjects of these questions, standing concerns their specified qualifications, admissibility retains the conditions of each use, and irreversibility keeps each result attributable to its own occurrence.

The proof-checker example in Section 6.5 has the same distinction. Supplying a missing premise produces a different proof performance; discovering that an unchanged proof already satisfied its rules changes knowledge or certification. Its independently defined inference relation and separately proved checker correctness make the conservation argument exact: warranted presentations of the same proof tree answer the same complete query alike. Agreement of their outputs is not a premise of their soundness proofs.

The retained-support result, Theorem B.11, applies when a later construction still requires the failed occurrence as a standing premise. A construction supplied with a fresh successful dependency is assessed on its actual support. This conserves the original dependency without prohibiting repair or further construction.

9.2. An identified physical record

Let a physical domain contain an identified record r and a claim comparing it with a later record r' . The record and its physical incidence already instantiate the kernel, independently of any descriptive or evidential use or adoption of AASC terminology. For the comparison to concern those records, the carrier, occurrences, question, and transformation under consideration must be identified. A coordinate change can preserve the same record; alteration of its substantive content presents a different comparison claim.

The conservation theorem compares answers only when the complete query has a determinate result under its substantive specification. An identified system need not have a unique outcome for every incompletely specified future experiment. A query about its admissible outcomes, a probability distribution, and a query about an actual realized record are different queries with different result types. Nothing in the proof replaces a set-valued or distribution-valued specification by a selected outcome. Determinacy must be established at the level of the answer actually claimed.

Nor does the historical result supply a thermodynamic arrow. A reversible physical evolution can be the subject of a determinate historical statement. Reversing the evolution does not turn its later occurrence

into the original one or replace the original conditions of the statement. Uncertainty about a record concerns access to its answer; an unsupported guess and a warranted answer have different standing for that use.

9.3. Role occupancy and diagnostic information

Suppose a target requires roles r_1, r_2 . Candidate x is eligible for r_1 , candidate y for r_2 , and a compatibility condition determines whether they can jointly occupy those roles. Candidate admission, eligibility, joint compatibility, and realization are distinct questions. Each completed admission assessment is binary, while the complete object retains the distinctions needed to ask those questions.

The role architecture must be fixed before candidates are compared.² A proof of unique compatible assignment does not itself produce the target unless a realization argument connects assignment and production. Likewise, a failed realization can yield a correct diagnostic report. Standing of the report as a report does not confer standing on the refused realization.

A status field that carries such diagnostic content is not erased merely because it is called a status field. The exact quotient keeps the content wherever the original observations require it. The status-only classification concerns admission for a specified use. The semantic theorem prevents a further discordant answer from acquiring authority at that unchanged use; it does not restrict the number of distinct objects, records, or mathematical properties in the domain.

9.4. What would challenge the argument?

The claims have different grounds and therefore different points of challenge. Merely renaming a function, distributing it across components, or deriving one implementation component from others does not remove its work. Once the original target commitments and their reconstruction are established, Theorem 3.6 excludes genuine deletion while retaining the same complete constructional job. A simpler implementation can be a substantive achievement without being an omission of the function it implements.

² For a downstream development of role-occupancy analysis, see [12]. No result from that analysis supplies a premise of the present necessity argument.

A foundational challenge must address the constitutive entailment in Proposition 2.4: whether an independently established non-degenerate determinate object or incidence necessarily incurs each stated requirement. The issue is whether the actual incidence requires the identity, qualification, original-occurrence, and answer-accountability distinctions, including when no one observes it. Showing only that these distinctions are useful to an investigator would leave that inference unestablished. Such a challenge can identify a gap in that inference or explain how the original constructional work remains possible without the allegedly necessary distinction. A reconstruction challenge instead concerns whether the work recovered from (A1)–(A4) has actually been shown to instantiate the four roles, without an added premise. A critic need not first accept the disputed requirement as part of “adequacy” and then attempt to omit what that definition already includes. These are challenges to the argument for necessity, not optional permissions governing whether the kernel applies to a determinate physical incidence. The reconstruction excludes functional omission while preserving the original complete job; it does not make the preceding constitutive reasoning immune to examination.

For the mathematical conservation theorem, a counterexample would require distinct answers with incidence under the same independently identified bearer, complete query, and determinate original result relation. The local proof excludes that combination. An advertised application has further obligations: its bearer identification, query preservation, result determinacy, and each original output’s warrant must actually be established. A claim to complete profile preservation must also cover the original observation family. Failure of one of these obligations is a genuine failure of that application, even though the conditional equality theorem remains valid. A richer raw code space with independently assigned reporting or reuse rules is not, without those connections, an established same-object counterexample. A changed complete bearer or question leaves that comparison, while every resulting non-degenerate determinate object remains within kernel necessity. A discordant report still claiming the original bearer and question instead fails its answer-role. Neither case supplies another standing state of the unchanged complete construction.

The lower-generator argument concerns finite acyclic certificates with their original-domain interpretations stated. Its authorization cut is a semantic point where warrant enters, not the syntactic cut rule; a cut-free proof in the latter sense can still supply warranted premises and licensing

inferences. A successful generator must expose where its output acquires the claimed warrant. A challenge can concern the classification of that point, the justification of its interpretation, or the inference that its successful application performs the necessary kernel work. Printing an authorization claim is insufficient, but marking a node as authoritative also does not prove it warranted. Infinitary or cyclic accounts lie beyond the finite lemma unless a further argument connects them to its certificate model. Their different syntax alone establishes neither a refutation nor a governance-free source of authority.

These distinctions also prevent an easy defence by redefinition. One may not add the desired result to the object's identity solely to force agreement, redefine the query after seeing conflicting outputs, remove an original observer, or count a timeout as a proved semantic refusal. A representation or generation claim must answer to the original conditions it purports to preserve.

10. Conclusion

The kernel expresses work already required by non-degenerate determinate objecthood: reference fixes what is concerned, standing identifies its qualification for a use, irreversibility retains the original occurrence and conditions, and admissibility governs reliance on those commitments together. An implementation may combine these functions, but genuinely deleting one loses the corresponding constructional requirement.

Representation conservation follows from an explicit local argument. Two independently warranted answers at the same determinate bearer and complete query agree. Quantifying over all original queries yields universal conservation of the governed profile, without requiring a finite observation syntax. A purported additional status cannot alter that profile while retaining the original bearer, complete question, and standing as its answer. It is either ineffective at the governance level, a change of bearer or constructional question, or an unsuccessful claim to the original answer-role.

Binary admission concerns the specified use, while faithful representation retains the substantive information needed to specify that use and its observations. The exact semantic and primitive quotient theorems connect these levels without discarding observed content. Together with authorization-cut analysis, full-profile uniqueness, symmetry restrictions,

and retained-support conservation, they give a fixed-domain exhaustion argument whose preservation claims are derived from independently warranted constructional facts.

The foundation therefore governs determinate mathematical and physical construction by necessity. Its strength lies in retaining the same object and the same complete question throughout the argument, exposing the standing of each proposed answer, and refusing to replace any of those obligations by a favorable label or an assumed agreement equation.

A. Representation and Implementation: Exact Interfaces

This appendix supplies the detailed implementation results for Section 8. It uses the reduction maps and primitive descent conditions already defined there. The primitive quotient, finite composition, and coverage arguments establish when a proposed interface realizes the semantic conservation theorem. They are not additional conditions for kernel governance to apply to a determinate object.

A.1. Exact quotient construction and composition

THEOREM A.1 (Exact primitive quotient criterion). *The following are equivalent:*

- (i) *every original primitive satisfies the descent conditions;*
- (ii) *a unique reduced partial algebra commutes with π and preserves and reflects primitive definedness;*
- (iii) *every generated finite strict typed term preserves definedness and reduced output under equal reduced input valuations.*

PROOF. Under (i), define $\bar{f}$ at a reduced tuple by applying f to any representative. Domain preservation makes definedness representative-independent; output preservation gives a unique reduced value. Surjectivity makes the resulting operation unique, giving (ii). Induct on a strict term for (iii). Variables have equal reductions and constants have identical values. At an application, argument definedness matches by induction. If an argument is undefined, both strict evaluations fail. Otherwise their reduced arguments agree, so the exact reduced primitive supplies matching definedness and reduced output. Conversely, applying (iii) to a primitive with distinct input variables gives (i). $\square$

LEMMA A.2 (Finite lazy conditionals). *The term-preservation conclusion extends to finite conditionals with ordinary observed Boolean tests.*

PROOF. By induction the tests have identical definedness and literal Boolean value. An undefined test makes both conditionals undefined. Otherwise both select the same branch, whose preservation follows by induction. The unselected branch is not evaluated. $\square$

COROLLARY A.3 (Full status reduction). *If every original observation is generated by the checked interface, primitive descent implies $s \approx_b t \iff \epsilon_b(s) = \epsilon_b(t)$. The same conclusion follows directly when all original*

observations meet the semantic conservation conditions under same-base, equal-polarity comparison.

PROOF. On the generated route, term preservation gives equal definedness and reduced outputs. Observed outputs are ordinary sorts, whose reduction is identity, so observations agree literally. On the semantic route, Theorem 6.3 gives the same agreement without a syntax restriction. In either case the admission observation proves the reverse implication; Theorem 8.5 identifies the quotient. $\square$

PROPOSITION A.4 (Observer-complete converse). *Suppose original observers detect primitive definedness, jointly distinguish ordinary outputs, and observe the retained base and polarity of status outputs. Suppose contexts allow precomposition by primitives and fixing their other arguments. If all observations preserve same-base, equal-polarity substitution, every primitive descends. Together with generated-observation coverage this is an equivalence.*

PROOF. Replace status inputs one at a time, leaving ordinary inputs identical. The precomposed definedness observer preserves the domain. A difference in an ordinary output would be separated by a precomposed observer, contradicting preservation. Base and polarity observers identify reduced status outputs. All intermediate tuples remain in the typed product; any application restriction is represented in the observed partial domain. The converse is Corollary A.3. $\square$

COROLLARY A.5 (A constructive generated interface). *An interface generated by ordinary partial operations, base and admission projections, status copying, and status constructors from ordinary inputs satisfies descent, provided the constructors' output base and polarity are fixed by those inputs. Composition and finite ordinary Boolean conditionals preserve this result. No further raw-status destructor is included.*

PROOF. Ordinary inputs are unchanged, so their partial operations preserve domains and values. Projections read retained coordinates; copying preserves the reduction. Constructor inputs are identical, so their domains agree and their specified reduced outputs agree. The primitive and conditional theorems then apply to every generated context. $\square$

These rules give an inspectable proof of preservation for a generated interface. Applying the result to an existing interface requires coverage of every original consumer. The semantic route proves incidence of

the original answers directly and extends beyond any such generated fragment.

A.2. Diagnostics and the scope of the quotient

A negative admission verdict and the record explaining it have different roles. Failed premises, incompatible boundary records, and unresolved obligations can be distinct substantive evidence. A standing report of failure reports that failure; it does not supply standing to the failed act. When an original observation reads such evidence, faithful comparison must retain the diagnostic object in its original data position or prove an exact representation of it.

For arbitrary raw codes, primitive descent and semantic validation are therefore substantive checks. A consumer copying an erased distinction into an ordinary result fails output descent; a consumer changing its application domain fails domain descent. If both are claimed to answer the same complete query about the same bearer, the conservation theorem shows that discordant outcomes cannot both be incident. Given a known incident answer, the discordant answer lacks standing in that role. This rejects its claimed authority without denying that the inscription is a separately determinate object or that another query may license a different use.

The repository's unrestricted `StatusInterface` exposes raw predicates for licensing, reports, and reuse. Its type alone provides no evidence that those predicates are incident answers about one fixed object and complete query. A separating assignment to those unrestricted fields consequently is not a counterexample to same-object conservation. The required semantic connection is exactly the identity and original-output validation used in Theorem 8.7.

Local binary reduction retains b . Hence it identifies the full reduced interface with a family of represented polarity fibres, not with two global object classes. For a domain-wide raw alphabet, a global binary verdict quotient additionally requires a compatible polarity map at every original typed occurrence, occurrence availability constant on its polarity classes, and preservation of every global observation. Values must be represented somewhere. Without these facts, differing bases or occurrence domains can distinguish values even when every individual admission question is binary. Conservation of an independently fixed bearer does not identify distinct bearers receiving the same verdict.

A.3. The resulting exhaustion statement

THEOREM A.6 (Same-object semantic and representational exhaustion). *Every incident presentation of an answer to one independently fixed, determinate bearer/query has the same semantic result. Discordant results cannot both retain that standing. For a proposed raw-interface reduction, exactly one of the following holds:*

- (i) *every primitive descends, yielding the unique exact reduced interface and preservation of its complete generated observations;*
- (ii) *a primitive has equal reduced inputs with different definedness or reduced outputs, precluding that exact reduction.*

The semantic conditions of Theorem 8.7 establish the first alternative for an interface realizing them.

PROOF. The universal semantic assertion and rejection consequence are Theorem 6.2 and Corollary 6.4. The primitive condition either holds throughout or has a counterinstance. Theorem A.1 proves existence and uniqueness in the first case and impossibility in the second. Semantic discharge supplies the local conditions, not a previously assumed global quotient. $\square$

The completed universal result is conservation for every actual standing answer at a fixed determinate object/query, together with unrestricted consumer invariance for its unique canonical status. Binary admission and faithful raw-code compression have the exact scopes established above. The proof neither exempts determinate objects from necessary governance nor confers standing on an arbitrary coding convention. It derives agreement from independently established identity, determinacy, and answer incidence.

B. Structural Consequences: Quotients, Symmetry, and Support

The necessary governance established above has concrete consequences for fixed observations, licensed operations, and retained dependencies. Three kinds of result must be distinguished. A semantic quotient identifies presentations indistinguishable by the fixed observations. Permutation symmetry describes material to which no individuating structure has yet been assigned. Dependency analysis determines how local checks support a historical construction. None supplies a new source of standing,

and none identifies arbitrary objects merely because both have positive standing.

B.1. The complete observation quotient

Fix the domain, target specifications, sorts, operation graphs, and standing-bearing observations of the construction under consideration. For a sort s , let X_s^+ be its standing presentations. Fixed observations include the relevant target and occurrence identifiers, licensed-use and reportability tests, and the observations made after licensed operations. Their codomains remain fixed during comparison. An operation can require more than standing of its inputs: its declared type and application conditions remain part of its graph.

DEFINITION B.1 (Complete contextual profile). A typed experiment on X_s^+ is a finite expression with one input position of sort s , formed from the fixed partial operations, arbitrary standing values used as fixed parameters, and a final fixed observation. Every standing value of the appropriate sort is available as such a parameter; no restricted list of named constants is assumed. Include definedness tests for such expressions. Undefined evaluation is recorded by a separate symbol $\perp$, distinct from every ordinary observation value. Close the experiment family under insertion into further finite typed contexts. Write

$$x \approx_s y \iff e(x) = e(y) \text{ for every experiment } e \text{ on sort } s.$$

The complete contextual profile is the family of these observation values. Put $Q_s = X_s^+ / \approx_s$, and let $q_s : X_s^+ \rightarrow Q_s$ be the quotient map.

Equality of all observations is an equivalence relation. This definition makes the information retained by the quotient explicit; it does not infer completeness from a shorter list of names. A proposed finite invariant tuple represents this quotient exactly only after both directions of the correspondence with $\approx_s$ have been established. In particular, equality of a target label, or agreement of standing polarity, alone does not supply that correspondence. The contextual profile records the fixed construction's semantics; it is not a new status classifier imposed on that construction. The conservation theorem of Section 6 supplies equality of these profiles for independently warranted presentations of the same bearer and complete questions. The quotient may also compare presentations whose identity

relation has not yet been established; semantic equivalence and a concrete proof of common bearer remain distinct assertions.

LEMMA B.2 (Congruence, including definedness). *Let*

$$\omega : X_{s_1}^+ \times \cdots \times X_{s_n}^+ \multimap X_t^+$$

be a fixed standing-preserving partial operation of finite arity. If $x_i \approx_{s_i} y_i$ for every i , then

$$\mathbf{x} \in \text{dom}(\omega) \iff \mathbf{y} \in \text{dom}(\omega).$$

When these conditions hold, $\omega(\mathbf{x}) \approx_t \omega(\mathbf{y})$. Fixed observed relations are likewise constant under replacement of each argument by an equivalent presentation.

PROOF. Replace the arguments one at a time. With the other arguments fixed, the question whether ω is defined is a typed experiment in the remaining position. Equivalence preserves its answer. To compare defined results, choose any experiment on the output sort and compose it with this application of ω . Context closure makes the composite an experiment on the replaced input, so its values agree, including any undefined value. Agreement for every output experiment is precisely output equivalence. Finite repetition proves the tuple statement. For a relation, its observed truth value with the other arguments fixed is itself an experiment; the same argument applies. $\square$

THEOREM B.3 (Semantic quotient and its universal property). *Every fixed operation in Lemma B.2 has a unique induced partial operation*

$$\bar{\omega} : Q_{s_1} \times \cdots \times Q_{s_n} \multimap Q_t$$

whose domain is the image of $\text{dom}(\omega)$ and whose value at a licensed tuple is $q_t(\omega(\mathbf{x}))$. For every map $F : X_s^+ \rightarrow Y$ constant on $\approx_s$ -classes, there is a unique map $\bar{F} : Q_s \rightarrow Y$ such that $F = \bar{F} \circ q_s$. Any other surjective classification with exactly these fibers is uniquely isomorphic to Q_s through its classification map.

PROOF. Lemma B.2 makes domain membership independent of the chosen representatives and gives equal output classes for all representatives. It therefore defines the induced partial operation. Surjectivity of the input quotients proves uniqueness. For the factorization, set $\bar{F}([x]) = F(x)$;

class constancy proves well-definedness, and surjectivity proves uniqueness. If $p : X_s^+ \rightarrow E$ is surjective with $p(x) = p(y)$ exactly when $x \approx_s y$, the induced map $[x] \mapsto p(x)$ is injective by the fiber condition and surjective by surjectivity of p . It is the unique bijection commuting with the two classification maps. $\square$

This is an exact semantic uniqueness result. It is independent of the number of classes: the separate status analysis determines when a particular status sort has a binary effective quotient. Quotient descent itself does not prove that all semantic distinctions reduce to a standing bit.

B.2. Application domains and the full interior

Let Ω_D be the fixed set of typed standing-preserving operations, including identity on each standing sort. Write $P_D = \coprod_s X_s^+$, retaining the sort tags. Define

$$\mathcal{A}_D = \coprod_{\omega \in \Omega_D} \text{dom}(\omega), \quad \text{ev}_D(\omega, \mathbf{x}) = \omega(\mathbf{x}).$$

An element of $\mathcal{A}_D$ records an operation together with a correctly typed tuple satisfying its fixed application conditions.

THEOREM B.4 (Total evaluation of licensed applications). *The evaluator $\text{ev}_D : \mathcal{A}_D \rightarrow P_D$ is total. Its domain is the unique largest application domain using exactly the fixed operation graphs. Every standing presentation occurs in its identity application. For composable unary operations,*

$$\text{dom}(g \circ f) = \{x \in \text{dom}(f) : f(x) \in \text{dom}(g)\}.$$

An extension preserving an old operation graph cannot give that operation a value at an old undefined input.

PROOF. Membership in $\mathcal{A}_D$ supplies the domain condition under which the recorded partial function has exactly one value. Every application using these graphs therefore belongs to $\mathcal{A}_D$, and its value is already determined. This proves totality on that domain and maximality. Identity has every point of its own sort in its domain. The composition formula follows by first evaluating f and then checking the domain of g . Giving an old undefined application a value would add a pair to its old graph, contradicting graph preservation. $\square$

Thus every licensed application is evaluable, while individual operations retain their partial domains. Neither self-application nor a particular binary combination follows from standing alone. The always-available application here is identity transport. These distinctions also apply to the quotient operations constructed above.

THEOREM B.5 (Full interior and generated closure). *The full admissible interior is uniquely P_D : a subset of the typed presentation universe is full precisely when it contains all and only the standing presentations. It is nonempty precisely when a standing presentation exists. For finitary operations in Ω_D , every seed set $B \subseteq P_D$ has a unique least closed extension $\text{Cl}_D(B) \subseteq P_D$. The full interior is the unique greatest closed admissible set, and $\text{Cl}_D(P_D) = P_D$.*

PROOF. Soundness and exhaustiveness of a full interior give equality with P_D pointwise; P_D itself meets both requirements. This also proves the nonemptiness assertion. For closure, let $B_0 = B$ and form B_{k+1} by adjoining every licensed output whose inputs lie in B_k . Include the values of licensed nullary operations. Standing preservation gives $B_k \subseteq P_D$ by induction. Put $B_\infty = \bigcup_{k < \omega} B_k$. Any finite tuple from B_∞ lies in one B_k , so its licensed output lies in B_{k+1} . Thus B_∞ is closed. Induction shows that every closed set containing B contains each B_k , proving leastness and uniqueness. Standing preservation makes P_D closed, and every admissible set is contained in it. Hence it is greatest and equals its own closure. $\square$

Full-interior uniqueness does not assert that every seed generates the whole interior. Different constructions may generate different closed subsets. Nor does the existence of a classification create a standing witness: nonemptiness requires an instantiated standing construction.

B.3. Semantic comparison and executable paths

PROPOSITION B.6 (Maximal comparison and generator completeness). *There is a canonical thin groupoid with objects X_s^+ and one comparison arrow $x \rightarrow y$ exactly when $x \approx_s y$. Its orbits are the semantic classes. It contains every thin comparison groupoid on the same objects whose arrows preserve all fixed observations. If Γ is a specified family of primitive redescription steps preserving those observations, its generated undirected components lie within the semantic classes. They equal those classes exactly when every equivalent pair is joined by a finite zigzag of Γ -steps.*

PROOF. Reflexivity, symmetry, and transitivity of $\approx_s$ give identities, inverses, and composition. Uniqueness of an arrow at fixed endpoints gives the groupoid laws. Its arrow criterion gives its orbits and maximality. Every primitive step and its reversal preserve equivalence, as does every finite composite; this proves component inclusion. Equality of components with classes is equivalent, by definition of a generated component, to the stated zigzag condition. $\square$

The maximal arrows certify semantic comparison; they do not assert an executable primitive instruction between every pair. Directed reachability requires its own directed paths and is stronger than undirected connectivity. A purported redescription that changes a fixed observation fails the comparison criterion. Its rejection as a redescription does not imply that its codomain, considered as a different object with its own target and history, is necessarily standing-negative. What fails is the asserted preservation of the original governed object.

B.4. The exact AMetric symmetry consequence

Before fixation, consider proposal material for which equality is available but no further individuating structure has been granted authority. Neutrality of such material means that every permutation is an admissible relabeling of the comparison problem. This symmetry is a property of the proposal interface, not of every later structured carrier. A fixed target or substantive compatibility relation can reduce the available symmetry. Here AMetric names the absence of an intrinsic ranking or grading among proposals before individuating structure is fixed.

DEFINITION B.7 (Equality-only proposal interface). Let P be a nonempty set with the full permutation group acting on it. A scalar observation $f : P^n \rightarrow Z$ has a fixed codomain, on which the group acts trivially. It is invariant if $f(\pi x_1, \dots, \pi x_n) = f(x_1, \dots, x_n)$ for every permutation π . For a tuple x , let $\text{Eq}(x)$ be the partition of $\{1, \dots, n\}$ given by $i \sim j$ precisely when $x_i = x_j$.

THEOREM B.8 (Finite-arity equality-pattern normal form). *For every finite n , the orbits of P^n under simultaneous permutation are exactly its realized equality partitions. Every invariant scalar observation factors uniquely through Eq . Conversely, every function on the realized equality partitions defines an invariant observation. When P is finite, a partition is realized exactly when it has at most $|P|$ blocks.*

PROOF. Permutations preserve equality and inequality, so an orbit has a fixed partition. Conversely, suppose two tuples have the same partition. The assignment $x_i \mapsto y_i$ is a well-defined bijection between their finite sets of entries, say A and B . On $U = A \cup B$, the still-unassigned domain $U \setminus A$ and range $U \setminus B$ have equal finite size. Choose a bijection between them, completing the assignment to a permutation of U , and extend by the identity outside U . This permutation sends the first tuple to the second. Equality partitions therefore classify the orbits, including the empty tuple when $n = 0$. An invariant observation is constant on each orbit, which supplies its unique factorization. The converse follows from preservation of equality. Finally, realizing a partition requires and suffices to assign a distinct element to each block. $\square$

COROLLARY B.9 (No individuation from unstructured symmetry). *Every invariant unary grade, score, or truth-valued classification on P is constant. If $|P| \geq 2$, there is no invariant strict total order and no input-independent selection of an element fixed by every permutation. An invariant real-valued metric is necessarily*

$$d(x, y) = \begin{cases} 0, & x = y, \\ c, & x \neq y, \end{cases} \quad c > 0,$$

when there are at least two elements. Such metrics exist for every $c > 0$.

PROOF. There is only one unary equality partition. Swapping two distinct elements reverses their order, contradicting an invariant strict total order; a transposition moving a selected point contradicts input-independent selection. There are exactly two pairwise equality patterns. The metric axioms fix the diagonal value at zero and the off-diagonal value at a positive constant. Conversely, the displayed function satisfies symmetry, separation, and the triangle inequality, and is invariant under all permutations. On a singleton the sole metric is zero. $\square$

The exact AMetric conclusion is absence of an intrinsic unequal unary assessment, privileged proposal, or relative nearness between distinct unindividuated proposals. Symmetry does not forbid equality tests, the discrete metric, constant quantities, or all operations. For example, an input-coordinate projection $p_i : P^n \rightarrow P$ is equivariant: $p_i(\pi \mathbf{x}) = \pi p_i(\mathbf{x})$. It is not a scalar invariant with a fixed codomain action and does not select a proposal independently of its inputs. Similarly, permutation symmetry fixes no particular positive metric scale. A nonconstant unary

assessment needs additional individuating data; the kernel requires those data and their role to be explicit. The symmetry theorem neither supplies such data nor grants them boundary authority. It also does not derive full permutation symmetry from determinacy alone: the equality-only character of this particular pre-fixational interface supplies the action used in the proof.

B.5. Historical support, trajectories, and reports

A construction's retained dependencies differ from its search history. An unsuccessful attempt may remain in a log without becoming a standing premise of a later construction. To track precisely which historical failures matter, represent each retained act by its original occurrence, target, specified use, and required predecessors. These records are fixed before evaluating a report of their support.

DEFINITION B.10 (Locally checked dependency record). Let H be a finite directed acyclic graph. An edge $u \rightarrow v$ declares that v requires u as a standing-bearing premise. Each vertex has a local check result $a(v) \in \{0, 1\}$ for its fixed occurrence and specified use. The check includes target and application conditions but does not silently replace the occurrence or remove a required premise. Define its support recursively in a topological order by

$$s_H(v) = a(v) \wedge \bigwedge_{u \rightarrow v} s_H(u),$$

with the empty conjunction equal to 1.

The local checks are required to be sound for the uses they check; a positive symbol unsupported by the relevant evidence is not such a check. For a concrete system, sound local checks and an accurate dependency record are the inputs to the following global preservation theorem. Their availability is an implementation obligation. For two encodings of the same local question, independent incidence of their results gives agreement by Section 6; local soundness itself must still be established from the original checking rules.

THEOREM B.11 (Exact support and retained failure). *For every vertex v ,*

$$s_H(v) = 1 \iff a(u) = 1 \text{ for every ancestor } u \text{ of } v, \text{ including } v.$$

A failed required ancestor therefore prevents positive support at every descendant retaining it. An extension preserving all old occurrence records, local results, and predecessor lists preserves every old support value. Any new vertex depending on an old unsupported vertex is unsupported.

PROOF. Induct in a topological ordering. A source has support equal to its local result. At a later vertex, the recurrence requires its own positive local result and positive support at every immediate predecessor. By induction, this is exactly positivity at every ancestor of those predecessors and at the vertex itself, which is the full ancestor set. In an extension, the recurrence for each old vertex has unchanged inputs, so the same induction preserves its value. An unsupported required predecessor forces a zero conjunct at its dependent vertex, and hence along every retained dependent path. $\square$

LEMMA B.12 (Local soundness gives global standing). *Let $J(v)$ be the standing-bearing assertion about the fixed occurrence and use of v . Suppose the local verification procedure establishes, for each vertex, the rule*

$$a(v) = 1 \wedge \bigwedge_{u \rightarrow v} J(u) \implies J(v).$$

Then $s_H(v) = 1$ implies $J(v)$ at every vertex.

PROOF. Induct in a topological ordering. Positive support supplies a positive local result and positive support at every predecessor. The induction hypothesis supplies their assertions $J(u)$; the verified local rule then gives $J(v)$. At a source there are no premise assertions to discharge. $\square$

This is a composition theorem for local verification, not an assumption of the global conclusion. A concrete checker must validate each local rule against the original use specification. Negative support establishes failure of this certificate to support the claimed record; the converse implication from $J(v)$ to positive support additionally requires completeness of the checks. No converse is used here.

COROLLARY B.13 (Finite trajectory certification). *For a finite nonempty dependency path $v_0 \rightarrow \dots \rightarrow v_n$, define coherence to mean $s_H(v_i) = 1$ at every index. Then coherence is equivalent to $s_H(v_n) = 1$. Coherent paths have coherent prefixes. Concatenating two paths at a shared*

endpoint yields a coherent path exactly when both paths are coherent. A path containing a failure remains incoherent under every extension preserving the old occurrence records, local results, and predecessor lists as in Theorem B.11.

PROOF. Every earlier path vertex is an ancestor of the terminal vertex. Positive terminal support therefore implies positive support at each earlier vertex by Theorem B.11; the converse is immediate. Prefix preservation and the composition equivalence follow because their vertices are respectively a subset and the union of the component vertices. A retained failure remains an unchanged witness in every such extension. $\square$

This gives the trajectory conclusion for paths whose edges are required dependencies. A more general licensed transition relation needs a separate proof that its standing failures propagate. Licensing the form of a step alone does not certify the standing of its outcome. A corrected construction may have positive support in a fresh dependency record; omitting a failed attempt there does not revise that attempt's original support or justify a positive report about the old retained record.

THEOREM B.14 (Reports from local evidence). *Consider the checking procedure for a positive report about v against the fixed record H that accepts exactly when the following checks pass: authenticity of the original occurrences, completeness of all required predecessor lists throughout the ancestor subgraph, and sound positive local evidence for every vertex there. Then acceptance implies $s_H(v) = 1$ and, under the verified local rules of Lemma B.12, the standing-bearing assertion $J(v)$. Conversely, suppose $s_H(v) = 1$, the ancestor record has verifiable evidence of its authenticity and complete required predecessor lists, and valid positive evidence for every required local check is available. Then these data give an accepting certificate for this checking procedure.*

PROOF. Completeness of the dependency check identifies exactly the required ancestors. Sound local evidence establishes $a(u) = 1$ for each of them. Theorem B.11 gives positive support at v , and Lemma B.12 gives $J(v)$ when its local soundness conditions hold. Conversely, positive support makes every recorded required ancestor positive. The stipulated authenticity and completeness evidence establishes that the finite record is the right record; the available positive local evidence then supplies the remaining checks. Since acceptance is exactly success of these checks, the resulting certificate is accepted. $\square$

This argument derives global support from local verification. It does not place global report soundness in a certificate field and then read that field as a proof. In particular, checking only the edges a report chooses to disclose would not establish the completeness condition. Duplicating a report changes neither the local checks nor its historical support.

PROPOSITION B.15 (Support preservation under lawful role reindexing). *Let $h : H \rightarrow H'$ be an isomorphism of finite dependency records preserving the identified occurrences, required uses, dependency edges, and local check results. If the records carry role assignments, require h also to preserve their role eligibility and joint compatibility data. Then $s_{H'}(h(v)) = s_H(v)$ for every vertex. Correspondingly transported complete local evidence preserves acceptance by the preceding report procedure.*

PROOF. The isomorphism identifies the predecessor sets and their local results. Induction in a topological ordering makes the two support recurrences equal. The same identification transports exactly the ancestor lists and local evidence inspected by the report checker, preserving its checks. $\square$

Role occupancy can therefore carry a preservation theorem once its eligibility, compatibility, and dependencies are fixed and verified. This result does not generate occupants, prove unique occupancy, or derive the kernel from an occupancy certificate that already uses it. Its direction is conservation of checked structure. Throughout this section, quotient comparison, licensed evaluation, symmetry, and support propagation preserve or expose the governance data of determinate construction; none supplies a repair channel for the original failed act.

C. Mechanized Necessity and Mathematical Support

The accompanying Lean 4.28.0 project checks the mathematical development of the necessity argument and its conservation, representation, and construction consequences. Its primitive necessity interface contains independently specified original subjects, complete uses, dependent witness types, and their incidence relation. It contains no field asserting (A1)–(A4), kernel governance, verdict agreement, or correctness of an arbitrary reader.

The necessity construction. The incidence module proves that an actual witness supplies positive qualification, that an actual qualification excludes a settled negative answer, and that unconditional transfer of arbitrary original grounds is possible exactly when their subjects agree. Original-incidence retention and non-substitution are proved separately from the classical construction of a total Boolean assessment. The latter uses excluded middle; it neither derives excluded middle from objecthood nor asserts a computable test or a unique physical outcome. The realization module proves the fibre criterion of Theorem 2.8, uniqueness on the realized image, reconstruction of original identity, and the omission, surplus, and erasure witnesses of Corollary 2.9. Its relational existence proof uses no choice of representative.

The assembled necessity theorem constructs a semantically exact assessment regime from the original relation, derives its represented four-role profile, and includes the universal exact-realization criterion for every proposed encoding. Its correspondence to (A1)–(A4) uses the accompanying identity, qualification, non-substitution and defect theorems together; the represented role record alone is not substituted for that chain. A separate bridge applies the finite authorization-cut theorem to target authority interpreted by the original relation. An actual original witness is extracted at the selected cut and supplies its constructed kernel realization. Kernel possession is not assumed as a property of an authority label. The first authoritative use need not be the identical query at the final node.

These proofs mechanize the stated necessity implications, including their nonvacuity and failure tests. Proposition 2.4 identifies the original semantic data with the actual identity-bearing incidence and explains why their retained functions are the four kernel roles. That interpretation is explicit and open to assessment; a compiler does not establish a physical interpretation merely from the name of a type. Equally, the necessity argument is not excluded from formalization because it is constitutive. Its mathematical development is checked, and the declaration map in the supplement distinguishes this development from empirical identification and from assertions about arbitrary incorrect representations.

Conservation and independently verified answers. The conservation modules prove equality from determinate original incidence and each answer’s separate original warrant. Query families may be arbitrary and result types may depend on the query; transport between equal query indices is

explicit. Complete optional outcomes identify semantic absence with absence of every original result, not with timeout or an uncompleted search. Canonical-status equality and total, partial, and dependent-consumer invariance are checked separately.

The proof-checker example is mechanized from its inference rules. It proves soundness and completeness of the recursive checker, uniqueness of its derivable result, exact negative-output semantics, and agreement between independently warranted readers of the same parsed bearer. Correctness is proved from the rules, not stored as a field. The formal instance uses natural-number atoms; the written argument permits any decidable atom alphabet.

Quotients, composition, and construction. The quotient modules construct the canonical quotient of arbitrary dependent observation profiles, its unique factorization, the actual profile-image realization, and the uniquely commuting isomorphism between exact classifiers. For finite heterogeneous operations, congruence is derived by replacing one argument at a time under independently interpreted original one-hole contexts. The quotient of input tuples is identified with the product of the sort quotients, and the partial operation descends uniquely with exactly its original domain image. The finite implementation language also has preservation proofs, including lazy conditionals.

Licensed finite-arity applications are evaluated in their fixed original graphs. Their generated closure is constructed, proved least, and identified with its finite stages; nullary operations are included. Equality-only interfaces have checked finite-pattern classification, invariant factorization, partition-realization criteria, binary normal forms, and strict-order and fixed-selector exclusions. Full carrier-permutation symmetry remains an explicit hypothesis of those particular results. The numerical metric specialization of the binary form retains its short written proof.

Dependencies and comparison. Finite semantic authorization is proved by selecting a globally minimal authoritative ancestor in an acyclic graph. Grounded support is generated from local checks and every required predecessor, then proved equivalent to the original ancestor conditions. Local rules yield global standing, and retained original dependencies preserve support. The semantic comparison relation is proved to be the maximal relation preserving the complete profile, with the identity, inverse, and composition laws of its thin groupoid. Finite generator zigzags

preserve the profile, and equality with the semantic comparison relation holds exactly when the specified generators connect every equal-profile pair. Counterexamples distinguish this completeness requirement from mere conservation and distinguish undirected comparison from directed execution. A node reindexing preserving and reflecting the original edges, together with the local eligibility and compatibility data, is proved to preserve and reflect ancestors, grounded support, and complete local evidence. Predecessor completeness is derived from the node equivalence and edge laws; support preservation is not assumed as a field.

Verification and scope. Lean checks proof terms and permits inspection of transitive axiom dependencies [8, 9]. The complete audit builds all project modules, inspects their constants, enumerates public theorem declarations by defining module, and requires an axiom report for every enumerated theorem. Missing, duplicate, or unexpected reports fail verification. No project axiom, unsafe proof, or placeholder is accepted. The permitted standard foundations are `propext`, `Classical.choice`, and `Quot.sound`, and constructive results are identified separately.

An explicit proof field remains a premise whenever a theorem uses it. A supplied soundness equation does not prove that an arbitrary proposed reader is sound. These constructions expose their independently specified relations and derive their realization properties. They preserve the direction of dependence: actual incidence fixes qualification, and a claimed presentation or use answers to those original conditions. A raw status interface whose reporting and reuse predicates are unconstrained by original incidence does not by itself establish a same-object realization. Its classical admitted-equivalent/not-admitted-equivalent split and the semantic conservation theorem have different stated premises.

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